

Micromechanics-inspired modeling of granular materials

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ABSTRACT

Micromechanics-based models are formulated at the particle scale with clear physics of anisotropy but pose computational efficiency challenges, particularly in finite element analysis. To leverage physics and reduce computational costs, this paper develops a micromechanics-inspired framework for modeling the mechanical behavior of granular materials with fabric anisotropy in two steps. In the first step, (1) the elastic compliance tensor is analytically derived from a micromechanical approach, directly incorporating fabric anisotropy into the elastic stiffness; (2) a transformed stress tensor, acting as an equivalent stress in a mapped isotropic configuration, is introduced into elastoplastic formulations to capture the coupled effect of fabric anisotropy on both elastic and plastic behaviors; and (3) a generalized strength criterion is formulated and integrated into the framework to consider the effect of intermediate principal stress. The elastic compliance tensor, transformed stress tensor, and generalized strength criterion are applied to a simple elastoplastic model and validated against experimental results. In the second step, A fabric-load coupling tensor accounting for the effect of non-proportionality between the loading direction and the fabric tensor is introduced. It is then used to introduce the combined stress tensor so that the framework is consistent with the anisotropic critical state theory (ACST). Any elastoplastic models incorporating a stress-dilatancy relationship can thus be extended to be consistent with ACST by replacing the stress tensor with the combined stress tensor in the flow rule. The SIMSAND model is selected as an example to demonstrate the implementation of the proposed framework. Validation across various loading paths, including constant stress ratio, triaxial shear, true triaxial shear, and torsional shear, confirms the model's capability to accurately capture the effect of fabric anisotropy on the mechanical behavior of granular materials. Furthermore, benchmarking results against the micromechanical model demonstrate the computational efficiency of the proposed framework.

1. Introduction

Granular materials often exhibit fabric anisotropy due to the gravitational deposition and loading processes. Various methods have been proposed in the literature to consider the effect of fabric anisotropy on the mechanical behavior of granular materials. Firstly, Pietruszczak and Mroz (2000) introduced a framework to generalize existing isotropic models by assuming that the yield functions of anisotropic models can be expressed as polynomials in the stress and fabric tensors. Secondly, Li and Dafalias (2004) introduced the fabric anisotropy variable A into existing isotropic elastoplastic models to account for fabric anisotropy. Numerous constitutive models were generalized to incorporate fabric anisotropy based on this concept. Examples include elastoplastic models (Wang et al., 2023; Ganesan and Mishra, 2024) and hypoplastic

models (Yang et al., 2020; Liao and Yang, 2021). Typically, these models introduced anisotropic parameters into elastoplastic formulations, including the critical state line, the dilatancy equation, the plastic modulus, the yield surface, and the plastic potential function. Thirdly, a transformed stress tensor (Tobita, 1988; Lou and Yoon, 2017; Shen et al., 2022) that transforms anisotropic materials into equivalent isotropic materials has been extensively employed in elastoplastic models to account for fabric anisotropy. Fourthly, Amorosi et al. (2026) proposed an anisotropic constitutive model for clays within a thermodynamic framework. Finally, to consider the effect of fabric anisotropy on the elastic behavior of granular materials, researchers introduced anisotropic elastic properties into elastoplastic models based on the frameworks of hyperelasticity (Amorosi et al., 2020; Rollo and

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Amorosi, 2022; Yu and Yang, 2025; Sun et al., 2025) or poroelasticity (Cheng and Yin, 2024, 2026).

Over the last several decades, micromechanical methods have been widely used to analyze the effect of fabric anisotropy on the mechanical behavior of granular materials. The micromechanical method estimates the macroscopic mechanical behaviors of the material from its microstructural response, thereby replacing a complex heterogeneous material with a homogeneous one. The macro–micro relationship yields the stress–strain relationship by averaging microscopic quantities statistically. Preliminary research on micromechanical methods dates back to the 19th century. Voigt (1889) proposed the assumption of uniform strain in heterogeneous materials, known as the Voigt assumption or the kinematic hypothesis. The kinematic hypothesis links the macroscopic strain of the material to the microscopic contact displacements between particles, assuming that the strain field of the particle assembly is uniform and that each particle moves in accordance with the strain imposed by the assembly. Under this assumption, the particle assembly is like a parallel spring system, and the failure condition of the system is that all springs fail simultaneously, so it corresponds to the upper bound solution of the material. In contrast, Reuss (1929) proposed the assumption of uniform stress in heterogeneous materials, known as the Reuss assumption or the static hypothesis. The static hypothesis assumes that the stress field of the particle assembly is uniform and that each particle experiences the same stress as the assembly. Under this assumption, the particle assembly is like a series spring system, and the failure condition of the system is that one spring fails, so it corresponds to the lower bound solution of the material. The kinematic and static hypotheses provide upper and lower bounds for the mechanical responses of heterogeneous materials. To incorporate plastic deformation into the micromechanical framework, researchers have proposed various micromechanical models based on the kinematic and static hypotheses. Nicot and his co-workers (Nicot et al., 2005; Nicot and Darve, 2007, 2011) developed the microdirectional model based on the kinematic hypothesis. Chang and Hicher (2005) proposed a micromechanical model based on the static hypothesis. This model has been widely used to simulate the mechanical behaviors of sands (Chang and Hicher, 2005), unsaturated soils (Hicher and Chang, 2007; Zhao et al., 2017), and clays (Chang et al., 2009; Yin et al., 2014). Besides, micromechanical methods have also been developed to simulate the mechanical behaviors of porous materials (Bignonnet et al., 2016; Chen and Pindera, 2020), rocks (Ulloa et al., 2022; Zhao et al., 2023; Zhu et al., 2026), and nano powders (Khoie et al., 2023). Among these micromechanical models, the Chang–Hicher (CH) model (Chang and Hicher, 2005) is one of the most widely used to investigate the effect of fabric anisotropy on the mechanical behavior of granular materials. The CH model adopts the static hypothesis for the anisotropic elastic stress–strain relationship and an elastoplastic contact law for plastic deformation. Since the real packing structure of the granular assembly is not fully considered in the CH model, the mechanical behaviors of granular materials, such as stress softening, stress dilatancy, and critical state behaviors, are reflected by the contact law.

While the aforementioned methods successfully address the effect of fabric anisotropy on the mechanical behaviors of granular materials, each method has its own drawbacks. On one hand, the introduction of the fabric anisotropy variable A into existing elastoplastic models (Li and Dafalias, 2004) ensures consistency with the anisotropic critical state theory (ACST, Li and Dafalias, 2012). Still, it relies on empirical assumptions regarding the effect of A on elastoplastic formulations, such as the yield surface and stress–dilatancy relationship. On the other hand, the approach of employing a transformed stress tensor to convert anisotropic materials into equivalent isotropic materials (Tobita, 1988; Yao et al., 2017; Zhang and Soga, 2025) has a micromechanical foundation; its compatibility with ACST remains uncertain. As for the micromechanical models (Chang and Hicher, 2005; Nicot et al., 2005), implementing them requires numerical integration over all possible

contact orientations to compute the stress tensor, which is computationally expensive. This is especially true for complex loading paths, which require numerous integration points to guarantee computational accuracy.

This paper proposes a micromechanics-inspired anisotropic elastoplastic framework for modeling the mechanical behavior of granular materials with fabric anisotropy in two steps. In the first step, the anisotropic elastic compliance tensor is derived analytically from the static hypothesis, which assumes that stress is uniformly distributed in the granular assembly. A transformed stress tensor and a generalized strength criterion are proposed and introduced into conventional elastoplastic formulations to establish an anisotropic elastoplastic framework. The proposed framework is applied to a simple elastoplastic model for validation. In the second step, the combined stress tensor is proposed and incorporated into the framework to ensure consistency with the anisotropic critical state theory. The proposed framework is applied to the SIMSAND model for validation.

2. Micromechanical approach based on the static hypothesis

Micromechanical models based on the static hypothesis have been widely employed to investigate the effect of fabric anisotropy on the mechanical behaviors of granular materials. The static hypothesis (Liao et al., 1997) assumes that stress is uniformly distributed in the granular assembly. It begins with the derivation of the contact forces from the stress tensor:

$$df_i^c = d\sigma_{ij} A_{jn} l_n^c \quad (1)$$

where f_i^c is the contact force, $d\sigma_{ij}$ is the stress increment, l_n^c is the branch vector, and A_{jn} is a fabric tensor that is defined as:

$$A_{jn} = \left(\frac{1}{V} \sum_{c=1}^N l_j^c l_n^c \right)^{-1} \quad (2)$$

where N/V is the number of contacts per unit volume. According to the definition, the fabric tensor A can be related to the widely used deviatoric fabric tensor F as:

$$A^{-1} = \frac{4Nr^2}{V} \mathbf{N} = \frac{4Nr^2}{V} \left(\frac{2}{15} \mathbf{F} + \frac{1}{3} \mathbf{I} \right) \quad (3)$$

where r is the particle radius, $\mathbf{I}$ is an identity matrix. Combined with the energy conservation principle and Love–Weber formula that averages the contact forces to obtain the stress tensor, the strain tensor can be calculated as:

$$dW = d\sigma_{ij} \epsilon_{ij} = \frac{1}{V} \sum_{c=1}^N df_i^c u_i^c \implies d\epsilon_{ij} = \frac{1}{V} \sum_{c=1}^N u_i^c l_n^c A_{jn} \quad (4)$$

where u_j^c is the contact displacement and W is the strain energy density. Combining Eqs. (1) and (4), the stress–strain relationship derived from the static hypothesis is given as (Liao et al., 1997):

$$d\epsilon_{ij} = S_{ijkl} d\sigma_{kl}, S_{ijkl} = \frac{1}{V} \sum_{c=1}^N \left(k_{jl}^c \right)^{-1} l_m^c l_n^c A_{mi} A_{nk} \quad (5)$$

where $k_{jl}^c = \text{diag}(k_n^c, k_r^c, k_r^c)$ is the contact stiffness matrix, $d\epsilon_{ij}$ is the strain increment, and S_{ijkl} is the compliance tensor.

To introduce plastic deformation into the micromechanical framework, Chang and Hicher (2005) proposed an elastoplastic contact law at the microscopic scale that is very similar to a classical macroscopic elastoplastic model. Thanks to the elastoplastic contact law, the model can capture key features of geomaterials, including pressure-dependent strength, dilatancy, and critical state behavior. The contact law was further extended to model the behavior of unsaturated soils (Hicher and Chang, 2007), clays (Chang et al., 2009), and sand–silt mixtures (Yin et al., 2014). At this point, it is evident that the micromechanical framework derived from the static hypothesis provides a general

platform for assessing the effect of fabric anisotropy on the mechanical behavior of geomaterials. At the same time, the elastoplastic contact law governs their plastic deformation.

3. Micromechanics-inspired anisotropic elastoplastic framework

The implementation of the aforementioned micromechanical model requires numerical integration over all possible contact orientations to compute the stress tensor, which is computationally expensive. Because an elastoplastic contact law governs the force–displacement relationship at each contact, the overall computational cost is significantly higher than that of conventional elastoplastic models. This is particularly true for complex loading paths, which require many integration points to ensure accuracy. As previously described, the micromechanical framework provides a general platform for incorporating fabric anisotropy, whereas the elastoplastic contact law governs the plastic deformation characteristics of geomaterials. Therefore, integrating the computationally expensive numerical integration of the micromechanical framework into conventional elastoplastic models is desirable to improve computational efficiency.

3.1. Anisotropic elasticity and transformed stress tensor

The determination of the elastic compliance tensor S_{ijkl} in Eq. (5) requires contact-level information, which is not available in conventional elastoplastic models. However, it is possible to theoretically integrate Eq. (5) to obtain an analytical expression of S_{ijkl} given the contact stiffness, branch vector, and a fabric tensor. Following the theory of representation for anisotropic materials, we assume that anisotropic materials can be equivalently expressed in a virtual isotropic configuration governed by σ^t , denoted as the transformed stress tensor. The mapping between the physical stress tensor σ and the transformed stress tensor σ^t is defined via the fabric tensor $\mathbf{F}$ to account for the structural anisotropy:

$$\sigma^t = \sigma \mathbf{A}^n(\mathbf{F}) = \sigma \mathbf{A}(\mathbf{A}^{\text{iso}})^{-1} = \sigma \left(\mathbf{I} + \frac{2}{5} \mathbf{F} \right)^{-1} \quad (6)$$

where $\mathbf{A}^n$ is a fabric tensor derived from the micromechanical static hypothesis, $\mathbf{A}^{\text{iso}} = 4Nr^2/(3V) \cdot \mathbf{I}$ is the fabric tensor of isotropic materials, and $\mathbf{I}$ is an identity matrix. Consequently, the strain tensor in the physical space can be derived as the following expression based on the energy conservation principle:

$$\epsilon = \frac{\partial W}{\partial \sigma} = \frac{\partial W}{\partial \sigma^t} \frac{\partial \sigma^t}{\partial \sigma} = \epsilon^t \mathbf{A}^n(\mathbf{F}) \quad (7)$$

Eq. (7) is not heuristic but a direct consequence of energy conservation under the assumed stress mapping. Substituting the transformed stress and strain tensors into Eqs. (1) and (4), the stress–strain relationship of the static hypothesis can be rewritten as:

$$\begin{aligned} df_i^c &= d\sigma_{ij}^c A_{jn} l_n^c = d\sigma_{ij}^t A_{jn}^{\text{iso}} l_n^c \\ d\epsilon_{ij}^t &= \frac{1}{V} \sum_{c=1}^N u_i^c l_n^c A_{jn} A_{jn}^{-1} A_{jn}^{\text{iso}} = \frac{1}{V} \sum_{c=1}^N u_i^c l_n^c A_{jn}^{\text{iso}} \end{aligned} \quad (8)$$

Eq. (8) shows that with the transformed stress tensor, the contact force can be computed using only the isotropic part of the fabric tensor. Therefore, the stress–strain relationship in Eq. (5) can be reformulated as:

$$d\epsilon_{ij}^t = S_{ijkl}^t d\sigma_{kl}^t, S_{ijkl}^t = \frac{1}{V} \sum_{c=1}^N \left(k_{jl}^c \right)^{-1} l_m^c l_n^c A_{mi}^{\text{iso}} A_{nk}^{\text{iso}} \quad (9)$$

The modified compliance tensor S_{ijkl}^t in Eq. (9) can be analytically integrated, as shown in Appendix. For cross-anisotropic materials, the

stress–strain relationship can be simplified as:

$$\begin{bmatrix} \epsilon_{11} \\ \epsilon_{22} \\ \epsilon_{33} \\ \epsilon_{12} \\ \epsilon_{31} \\ \epsilon_{23} \end{bmatrix} = \begin{bmatrix} \frac{1}{E_h} & -\frac{\nu_{hh}}{E_h} & -\frac{\nu_{vh}}{E_v} & 0 & 0 & 0 \\ -\frac{\nu_{hh}}{E_h} & \frac{1}{E_h} & -\frac{\nu_{vh}}{E_v} & 0 & 0 & 0 \\ -\frac{\nu_{vh}}{E_v} & -\frac{\nu_{vh}}{E_v} & \frac{1}{E_v} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{2G_{hh}} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{1}{2G_{vh}} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{2G_{hv}} \end{bmatrix} \begin{bmatrix} \sigma_{11} \\ \sigma_{22} \\ \sigma_{33} \\ \sigma_{12} \\ \sigma_{31} \\ \sigma_{23} \end{bmatrix} \quad (10)$$

where elastic properties are expressed as:

$$\begin{aligned} E_h &= \frac{(1 - \Delta/5)^2}{1 - c_1 \Delta/2} E, E_v = \frac{(1 + 2\Delta/5)^2}{1 + c_1 \Delta} E, c_1 = \frac{2(7 - 9\nu)}{35} \\ \nu_{vh} &= \frac{(1 + \Delta/7)(1 + 2\Delta/5)\nu}{(1 - \Delta/5)(1 + c_1 \Delta)}, \nu_{hv} = \frac{(1 + \Delta/7)(1 - \Delta/5)\nu}{(1 + 2\Delta/5)(1 - c_1 \Delta/2)}, \\ \nu_{hh} &= \frac{(1 - 2\Delta/7)\nu}{1 - c_1 \Delta/2} \\ G_{hv} &= G_{vh} = \frac{(1 - \Delta/5)(1 + 2\Delta/5)G}{(1 + \Delta/10)}, G_{hh} = (1 - \Delta/5)G \end{aligned} \quad (11)$$

To validate the analytical solutions for elastic properties derived from the micromechanical framework, numerical integrations of Eq. (5) are performed using the method proposed by Bažant and Oh (1986) with 74 integration points. Assuming the deviatoric fabric tensor for an orthotropic material is defined as:

$$\mathbf{F} = \begin{bmatrix} -0.3 & 0 & 0 \\ 0 & -0.1 & 0 \\ 0 & 0 & 0.4 \end{bmatrix} \quad (12)$$

To validate the analytical solutions in general conditions, the fabric tensor is rotated by 45° around the y -axis as:

$$\begin{aligned} \mathbf{F}' &= \mathbf{R} \mathbf{F} \mathbf{R}^{-1} = \begin{bmatrix} 0.05 & 0 & 0.35 \\ 0 & -0.1 & 0 \\ 0.35 & 0 & 0.05 \end{bmatrix}, \mathbf{R} = \begin{bmatrix} \cos \theta & 0 & \sin \theta \\ 0 & 1 & 0 \\ -\sin \theta & 0 & \cos \theta \end{bmatrix} \\ &= \begin{bmatrix} \sqrt{2}/2 & 0 & \sqrt{2}/2 \\ 0 & 1 & 0 \\ -\sqrt{2}/2 & 0 & \sqrt{2}/2 \end{bmatrix} \end{aligned} \quad (13)$$

Based on the rotated fabric tensor, the analytical and numerical solutions of Young's modulus (E_{33}/E), shear modulus (G_{31}/G), and Poisson's ratio (ν_{31}/ν) are shown in Fig. 1. The figure shows that the analytical solutions for elastic properties derived from the micromechanical framework agree well with the numerical solutions, demonstrating their correctness.

3.2. Anisotropic elastoplastic framework

According to Eq. (8), the contact force derived from the static hypothesis within the micromechanical framework can be computed using an isotropic fabric tensor based on the transformed stress tensor. This finding suggests that fabric anisotropy can be incorporated into conventional elastoplastic models by using a transformed stress tensor, similar to the approach used in anisotropic elasticity. Therefore, an anisotropic elastoplastic framework can be established by replacing the stress tensor in conventional elastoplastic models with the transformed stress tensor defined in Eq. (6). It should be noted that the symmetric part of the transformed stress tensor is used in the elastoplastic formulations. The relative difference between the deviatoric stress of the transformed stress tensor and that of its symmetric part is found to be small, with a maximum value below 4%. This indicates that considering only the symmetric part of the transformed stress tensor has a negligible effect on the elastoplastic response. For simplicity, the symmetric part of the transformed stress tensor is still denoted as σ^t in the following

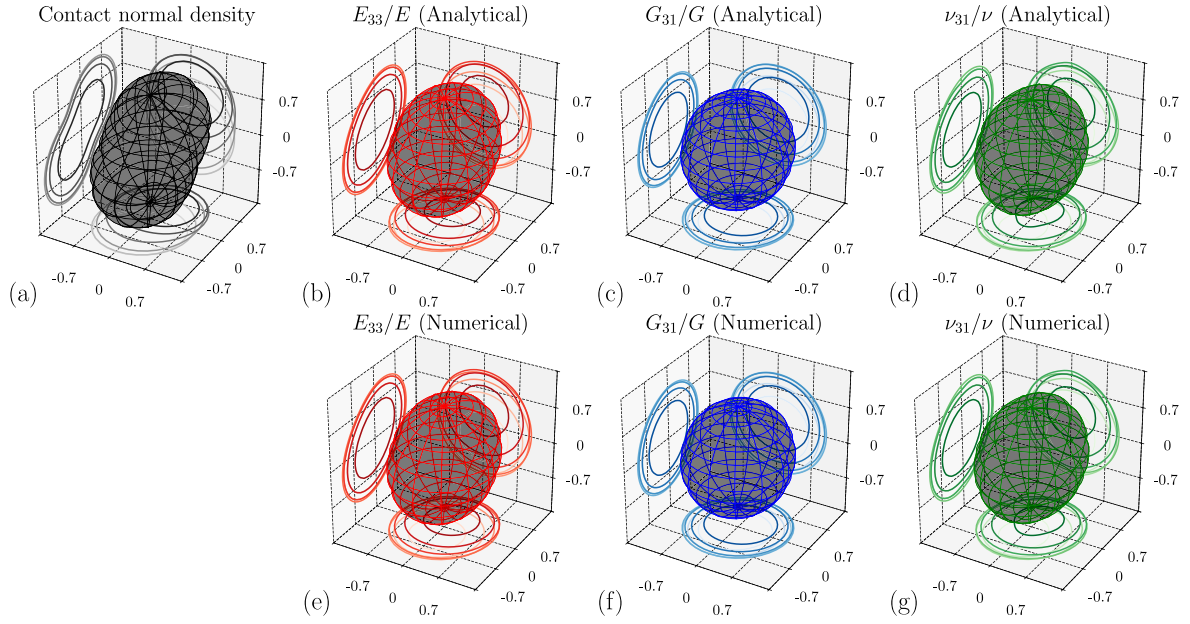


Fig. 1. Comparison between analytical and numerical solutions of elastic properties: (a) contact normal density; (b)–(d) analytical solutions of E_{33}/E , G_{31}/G and ν_{31}/ν ; (e)–(g) numerical solutions of E_{33}/E , G_{31}/G and ν_{31}/ν .

sections. Indeed, the transformed stress tensor is equivalent to that used in previous studies (Tobita, 1988; Yao et al., 2017; Zhang and Soga, 2025). The novelty of our work lies in the consideration of the anisotropic elasticity which is derived from micromechanical approach that inspires the proposed framework, guaranteeing the consistency of the elastic and plastic behaviors of granular materials with fabric anisotropy.

Assuming the yield function and potential function of a conventional elastoplastic model are defined as $f(\sigma, \kappa) = 0$ and $g(\sigma)$ respectively, the elastoplastic stiffness tensor of the model can be expressed as:

$$D_{ijkl}^{\text{ep}} = D_{ijkl} - \frac{D_{ijmn} \frac{\partial g}{\partial \sigma_{mn}} \frac{\partial f}{\partial \sigma_{pq}} D_{pqkl}}{\frac{\partial f}{\partial \sigma_{mn}} D_{mnpq} \frac{\partial g}{\partial \sigma_{pq}} - \frac{\partial f}{\partial \kappa} \frac{\partial \kappa}{\partial \epsilon_{mn}^p} \frac{\partial g}{\partial \sigma_{mn}}} \quad (14)$$

where D_{ijkl} is the elastic stiffness tensor, ϵ_{ij}^p is the plastic strain tensor, κ is the hardening variable.

To generalize the elastoplastic model to account for fabric anisotropy, the transformed stress tensor defined in Eq. (6) is introduced into the yield function and potential function, resulting in $f(\sigma^t, \kappa) = 0$ and $g(\sigma^t)$. Consequently, the derivatives of the yield and potential functions with respect to the original stress tensor can be derived as:

$$\frac{\partial f}{\partial \sigma_{ij}} = \frac{\partial f}{\partial \sigma_{kl}^t} \frac{\partial \sigma_{kl}^t}{\partial \sigma_{ij}}, \quad \frac{\partial g}{\partial \sigma_{ij}} = \frac{\partial g}{\partial \sigma_{kl}^t} \frac{\partial \sigma_{kl}^t}{\partial \sigma_{ij}} \quad (15)$$

where $\partial f / \partial \sigma_{ij}^t$ and $\partial g / \partial \sigma_{ij}^t$ are similar to those in conventional elastoplastic models except that the stress tensor is replaced by the transformed stress tensor, $\partial \sigma_{kl}^t / \partial \sigma_{ij}$ is the derivative of the transformed stress tensor with respect to the original stress tensor. Therefore, the elastoplastic stiffness tensor of the anisotropic elastoplastic framework can be calculated as:

$$D_{ijkl}^{\text{ep}} = D_{ijkl} - \frac{D_{ijmn} \frac{\partial g}{\partial \sigma_{uv}^t} \frac{\partial \sigma_{uv}^t}{\partial \sigma_{mn}} \frac{\partial f}{\partial \sigma_{st}^t} \frac{\partial \sigma_{st}^t}{\partial \sigma_{pq}} D_{pqkl}}{\frac{\partial f}{\partial \sigma_{st}^t} \frac{\partial \sigma_{st}^t}{\partial \sigma_{mn}} D_{mnpq} \frac{\partial g}{\partial \sigma_{uv}^t} \frac{\partial \sigma_{uv}^t}{\partial \sigma_{pq}} - \frac{\partial f}{\partial \kappa} \frac{\partial \kappa}{\partial \epsilon_{mn}^p} \frac{\partial g}{\partial \sigma_{uv}^t} \frac{\partial \sigma_{uv}^t}{\partial \sigma_{mn}}} \quad (16)$$

3.3. Fabric tensor of granular materials

Granular materials typically exhibit a transversely isotropic fabric resulting from gravitational deposition. The deviatoric fabric tensor for such materials is expressed as:

$$\mathbf{F} = \begin{bmatrix} \Delta & 0 & 0 \\ 0 & -\frac{\Delta}{2} & 0 \\ 0 & 0 & -\frac{\Delta}{2} \end{bmatrix} \quad (17)$$

where Δ is a parameter controlling the degree of fabric anisotropy. When subjected to non-coaxial loading relative to the initial fabric orientation, as illustrated in Fig. 2, the fabric tensor must be rotated accordingly to capture the non-coaxial behavior of granular materials:

$$\mathbf{F}' = \mathbf{R} \mathbf{F} \mathbf{R}^{-1} \quad (18)$$

$$\mathbf{R} = \begin{bmatrix} \cos \alpha & 0 & \sin \alpha \\ 0 & 1 & 0 \\ -\sin \alpha & 0 & \cos \alpha \end{bmatrix}$$

where $\mathbf{R}$ is a rotation matrix.

The evolution of fabric anisotropy during shearing is crucial for capturing the anisotropic mechanical response of granular materials. According to Li and Dafalias (2012), the fabric evolves towards a steady state at the critical state under continuous shearing. Consequently, this study adopts a fabric evolution law based on the work of Li and Dafalias (2012):

$$d\mathbf{F} = \langle \lambda \rangle k_f (\mathbf{n} - \mathbf{F}) \quad (19)$$

where $\langle \lambda \rangle$ is the norm of the plastic strain increment, $\mathbf{n}$ is the loading direction tensor, k_f is a model parameter controlling the rate of fabric evolution. To describe the relative orientation between the fabric tensor and the loading direction, the fabric anisotropy variable A is defined as:

$$A = \mathbf{F} : \mathbf{n} = F \mathbf{n}_F : \mathbf{n} = F N \quad (20)$$

where F is the norm of the fabric tensor, and N is the norm of the loading direction tensor. When A approaches 1, the fabric tensor aligns with the loading direction, indicating that the material's fabric is aligned with the loading direction. Based on DEM simulations (Fu

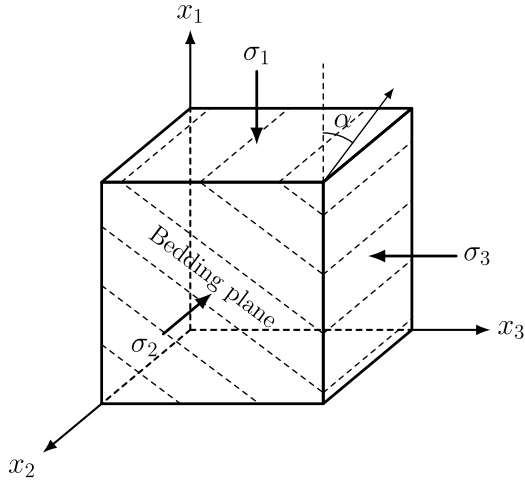


Fig. 2. Relationship between loading direction and material fabric.

and Dafalias, 2015) and experimental investigations (Mitaritonna et al., 2014), the degree of fabric anisotropy at the critical state is found to be dependent on the stress ratio. Therefore, Yu and Yang (2025) proposed a fabric evolution law that relates the degree of fabric anisotropy at the critical state to the stress ratio as:

$$\begin{aligned} d\mathbf{F} &= \langle \lambda \rangle k_f \{ [d_1 (\eta/M - 1) + 1] \mathbf{n} - \mathbf{F} \} \\ \mathbf{n} &= \mathbf{s} / \|\mathbf{s}\|, d_1 = c_1 \exp [c_2 (\eta^2 - c_3^2)] \end{aligned} \quad (21)$$

where $\eta = q/p$ is the stress ratio, $\mathbf{s} = \boldsymbol{\sigma} - p\mathbf{I}$ is the deviatoric stress tensor, M is the critical stress ratio, and k_f , c_1 , c_2 , and c_3 are model parameters. When plastic strain is large enough, the norm of the fabric tensor F approaches $d_1 (|\eta|/M - 1) + 1$. The calibration of the model parameters k_f , c_1 , c_2 , and c_3 can be found in Yu and Yang (2025) and will not be discussed in this paper.

3.4. Generalized strength criterion and integration

The proposed anisotropic elastoplastic framework can be applied to generalize existing elastoplastic models of granular materials to account for fabric anisotropy. These models typically employ the Mohr–Coulomb strength criterion to describe the strength characteristics of granular materials. Researchers have proposed various methods to generalize the Mohr–Coulomb model to account for the effect of intermediate principal stress. For example, Yao et al. (2004) proposed a generalized strength criterion based on the SMP criterion (Matsuoka and Nakai, 1974) and the Von Mises criterion; Mortara (2010) proposed a method in which a scalar function of Lode angle was used to modify the shape of the failure criterion in Mortara (2008). The family of $g(\theta)$ method for modifying the strength in the principal stress space is more widely used to generalize the Mohr–Coulomb model and a plenty of $g(\theta)$ functions (Argyris et al., 1974; Stake, 1982; Sheng et al., 2000) have been proposed. This paper adopts a more general method proposed by Wang et al. (2026) that can be used to generalize existing strength criteria within the framework of the Mohr–Coulomb model. According to Wang et al. (2026), the critical stress ratio M^t of the Mohr–Coulomb model can be expressed in terms of critical friction angle ϕ_c and intermediate principal stress ratio b^t as:

$$M^t = \frac{6\sqrt{(b^t)^2 - b^t + 1} \sin \phi_c}{3 + (2b^t - 1) \sin \phi_c} \quad (22)$$

where b^t is defined as $b^t = (\sigma_2^t - \sigma_3^t) / (\sigma_1^t - \sigma_3^t)$, with σ_1^t , σ_2^t , and σ_3^t being the major, intermediate, and minor principal values of the transformed stress tensor.

Although the proposed anisotropic elastoplastic framework can inherently capture the effect of fabric anisotropy on the strength of

granular materials under general stress states, the effect of intermediate principal stress is not explicitly considered because it reduces to the Mohr–Coulomb model at the critical state. Wang et al. (2026) proposed a method to generalize the Mohr–Coulomb model to account for the effect of the intermediate principal stress by using a Lode angle-dependent friction angle, which can be applied to generalize existing strength criteria. According to Wang et al. (2026), a generalized strength criterion can be established by replacing the friction angle ϕ_c in Eq. (22) with the following transformed friction angle ϕ_c^t :

$$\begin{aligned} \sin \phi_c^t &= \omega_{MC} \sin \phi_c + \omega_{Mises} \sin \phi_{Mises} + \omega_{Tresca} \sin \phi_{Tresca} + \omega_{SMP} \sin \phi_{SMP} \\ \text{with } \sin \phi_{Mises} &= \frac{3 \sin \phi_c}{3\sqrt{(b^t)^2 - b^t + 1} - \left(2b^t + \sqrt{(b^t)^2 - b^t + 1} - 1\right) \sin \phi_c} \\ \sin \phi_{Tresca} &= \frac{3 \sin \phi_c}{3 - 2b^t \sin \phi_c} \\ \sin \phi_{SMP} &= \frac{\tan \phi_{SMP}}{\sqrt{1 + \tan^2 \phi_{SMP}}} \text{ with} \\ \tan \phi_{SMP} &= \sqrt{\frac{2}{1 + \frac{(1 - b^t)^2 + (b^t)^2 R^t}{1 - b^t + b^t R^t}}} \tan \phi_c \end{aligned} \quad (23)$$

where $R^t = \sigma_1^t / \sigma_3^t$, ϕ_{Mises} , ϕ_{Tresca} , and ϕ_{SMP} are equivalent friction angles for the Mises, Tresca, and SMP criteria, ω_{MC} , ω_{Mises} , ω_{Tresca} , and ω_{SMP} are weights for each criterion. The weighting coefficients are only introduced to demonstrate the capability of the proposed framework to generalize existing strength criteria to account for the effect of intermediate principal stress. For granular soils, the SMP and Von Mises criteria are often combined to form a generalized strength criterion, and only one weighting coefficient is required to quantify the contribution of the two criteria.

A parametric study is conducted to demonstrate the effects of anisotropy degree and loading direction on the strength behaviors of granular materials. Fig. 3 presents strength envelopes for the generalized strength criterion under various degrees of anisotropy and loading directions. The results indicate that the effects of anisotropy degree and loading direction on strength are consistent for different criteria. The strength envelopes demonstrate that fabric anisotropy significantly influences the strength of granular materials. A higher degree of fabric anisotropy results in greater strength anisotropy, while a change in the loading direction results in the rotation of strength envelopes. Furthermore, the strength envelopes are nearly identical to those presented by Wang et al. (2026), in which the strength criteria are derived analytically from the micromechanical model. This agreement confirms the validity of the proposed anisotropic framework for considering the effect of fabric anisotropy on the strength anisotropy of granular materials.

To this end, an anisotropic elastoplastic framework has been established. This framework can be used to generalize conventional elastoplastic models to account for the effects of fabric anisotropy (via the transformed stress tensor σ^t) and the intermediate principal stress (via the transformed friction angle ϕ_c^t) on the mechanical behavior of granular materials.

4. Application to a simple elastoplastic model

This section applies the proposed framework to a simple elastoplastic model, a simplified version of the SIMSAND model (Yin et al., 2018) that does not consider the critical state concept.

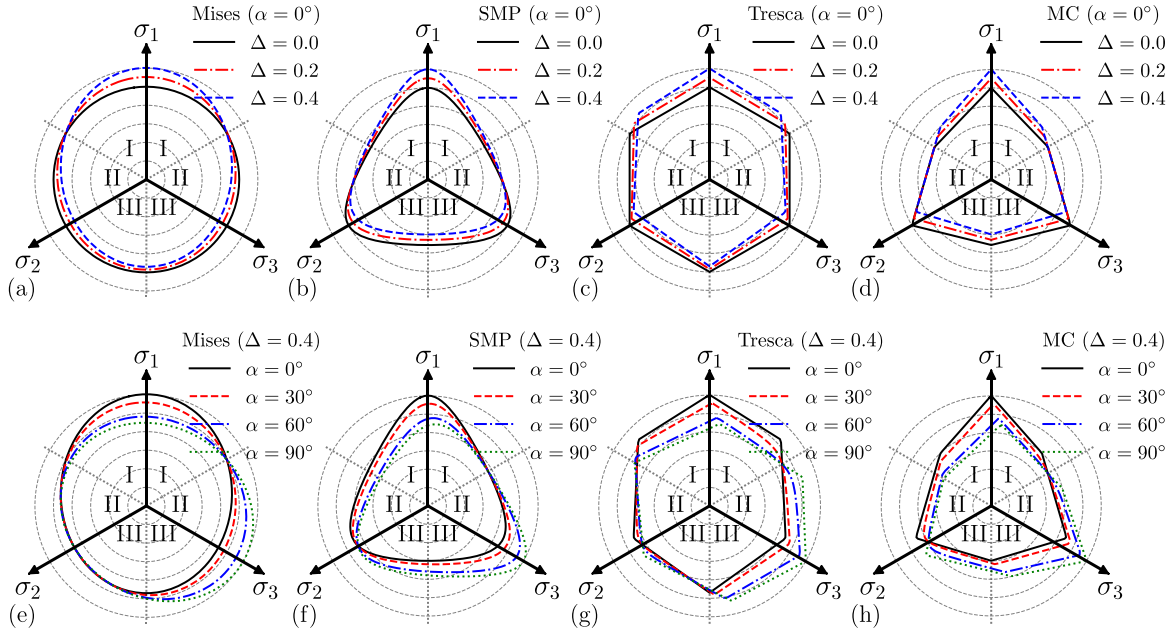


Fig. 3. Effect of anisotropy degree and loading direction on strength anisotropy: (a)–(d) effect of anisotropy degree; (e)–(h) effect of loading direction.

4.1. The simple elastoplastic model

The yield function of the simple elastoplastic model is defined as:

$$f(\sigma^t) = \frac{q^t}{p^t} - \frac{M_{pt}^t \varepsilon_d^p}{k_p + \varepsilon_d^p} \quad (24)$$

where p^t and q^t are the mean effective stress and deviatoric stress based on the transformed stress tensor, $M_{pt}^t = M^t$ is the peak stress ratio, ε_d^p is the deviatoric plastic strain, and k_p is a hardening parameter. The stress-dilatancy relationship of the model is given as:

$$D = A_d \left(M_{pt}^t - \frac{q^t}{p^t} \right), \frac{\partial g}{\partial p^t} = D, \frac{\partial g}{\partial q^t} = 1 \quad (25)$$

where A_d is a dilatancy parameter, M_{pt}^t is the phase-transformation stress ratio where no volumetric plastic strain occurs. M_{pt}^t can be expressed in terms of the phase-transformation friction angle ϕ_{pt} :

$$M_{pt}^t = \frac{6\sqrt{(b^t)^2 - b^t + 1} \sin \phi_{pt}}{3 + (2b^t - 1) \sin \phi_{pt}} \quad (26)$$

Under isotropic conditions, Eq. (25) reduces to the stress-dilatancy relationship of the isotropic model:

$$\frac{d\varepsilon_v^p}{d\varepsilon_d^p} = A_d \left(M_{pt}^t - \frac{q^t}{p^t} \right) \quad (27)$$

where ε_v^p is the plastic volumetric strain, and ε_d^p is the plastic shear strain. To consider the effects of void ratio and stress level on the elastic behavior of granular materials, the following shear modulus is adopted in the simple elastoplastic model:

$$G = G_0 \frac{(2.97 - e)^2}{1 + e} \left(\frac{p^t}{p_a} \right)^n \quad (28)$$

where G_0 is the reference shear modulus, and n is a model parameter.

4.2. Calibration of model parameters

The anisotropic elastoplastic model requires several parameters to be calibrated using experimental data. Detailed calibration procedures are presented below:

- The elastic parameters (G_0 , ν) and initial degree of fabric anisotropy (Δ) can be determined based on measured elastic properties. Based on Eq. (11), E , G , and Δ can be calculated as:

$$\Delta = \frac{10(1 - G_{hh}/G_{hv})}{4G_{hh}/G_{hv} - 1}, E = \frac{(1 + c_1\Delta)E_v}{(1 + 2\Delta/5)^2} = \frac{(1 - c_1\Delta/2)E_h}{(1 - \Delta/5)^2} \quad (29)$$

$$G = G_0 \frac{(2.97 - e)^2}{1 + e} \left(\frac{p^t}{p_a} \right)^n = \frac{G_{hh}}{1 - \Delta/5} = \frac{(1 + \Delta/10)G_{hv}}{(1 - \Delta/5)(1 + 2\Delta/5)}$$

The Poisson's ratio ν can be determined from the relationship between E and G as $\nu = E/(2G) - 1$.

- The critical friction angle (ϕ_c) can be determined from the stress ratio at the critical state. According to Wang et al. (2026), the critical stress ratios under conventional triaxial compression and triaxial extension loadings can be expressed as:

$$M_{TC} = \frac{(6 + 3\Delta_f/5) \sin \phi_c + 9\Delta_f/5}{3 - (1 - 4\Delta_f/5) \sin \phi_c} \quad (30)$$

$$M_{TE} = \frac{(6 + 3\Delta_f/5) \sin \phi_c - 9\Delta_f/5}{3 + (1 - 4\Delta_f/5) \sin \phi_c}$$

where M_{TC} and M_{TE} are the critical stress ratios under triaxial compression and triaxial extension loadings, respectively. Δ_f is the degree of fabric anisotropy at the critical state, which is generally related to the loading direction $\mathbf{n}$. Based on Eq. (30), ϕ_c can be determined from the measured critical stress ratios as:

$$\sin \phi_c = \frac{3M_{TC} - 9\Delta_f/5}{6 + 3\Delta_f/5 + (1 - 4\Delta_f/5) M_{TC}} \quad (31)$$

$$\sin \phi_c = \frac{3M_{TE} + 9\Delta_f/5}{6 + 3\Delta_f/5 - (1 - 4\Delta_f/5) M_{TE}}$$

- The dilatancy parameter (A_d and ϕ_{pt}) can be calibrated using $\varepsilon_q - \varepsilon_v$ curves of drained triaxial tests, which are generally the same as conventional elastoplastic models.
- The fabric evolution parameters (k_f , c_1 , c_2 , c_3) can be calibrated from the evolution of fabric anisotropy in triaxial compression tests and constant-stress-ratio tests. A detailed calibration procedure is described in Yu and Yang (2025).
- Other parameters (n , k_p) can be determined by fitting the shear stress-strain curves of drained and undrained triaxial tests.

Table 1Summary of model parameters for the simple elastoplastic model (unit: G_0 in MPa, ϕ_c and ϕ_{pt} in degrees).

Soil type	G_0	ν	n	ϕ_c	ϕ_{pt}	k_p	A_d	Δ	k_f	c_1	c_2	c_3
Lucera clay	4.2	0.2	1.0	32.8	32.8	1e-7	1.45	-0.29	62.4	2.81	1.86	0.66
Cambria sand	10.4	0.2	0.56	35.0	29.3	9e-4	1.41	0.2	3.44	0	0	0

Table 2

Summary of model parameters for true triaxial tests.

Soil	Reference	ϕ_c (°)	F_{11}	F_{22}	F_{33}	ω_{mises}	ω_{SMP}
Cambria sand	Ochiai and Lade (1983)	39.76	0.106	0.020	-0.126	0.271	0.729
San Francisco Bay mud	Kirkgard and Lade (1993)	38.64	0.339	-0.095	-0.244	0.134	0.866
Santa Monica Beach sand	Abelev and Lade (2004)	45.60	0.324	-0.060	-0.264	0.179	0.821
Glass beads	Haruyama (1981)	28.68	0.285	-0.112	-0.173	0.446	0.554

In the following sections, a series of experiments is simulated to verify the performance of the proposed anisotropic elastoplastic model. Table 1 lists the model parameters used in simulations. The fabric evolution law in Eq. (21) is adopted in simulations of anisotropic elastic properties under constant stress ratios. For available tests with only one single stress path (e.g. only drained triaxial tests), Eq. (21) reduces to Eq. (19) due to lack of data and therefore, c_1 , c_2 , and c_3 are set to zero.

4.3. Simulations of anisotropic elastic properties

Experimental results from Mitaritonna et al. (2014) for Lucera clay are used to validate the derived anisotropic elastic properties shown in Eq. (10). The proposed anisotropic elastoplastic model, combined with the fabric evolution law in Eq. (21), is then used to simulate the experimental results from Mitaritonna et al. (2014), in which Lucera clay was loaded at constant stress ratios of 0, 0.3, 0.6, and 0.8. Fig. 4 compares the simulations and experimental results for the anisotropic elastic properties of Lucera clay. The figure demonstrates that the anisotropic elastoplastic model accurately captures the evolution of the shear stiffness ratio (G_{hh}/G_{hv}) under different stress paths, as shown in Fig. 4(b). For the case of $\eta = 0.8$, the shear stiffness anisotropy becomes significant during loading, with G_{hh}/G_{hv} increasing from 1.1 to approximately 1.2. In contrast, for the cases of $\eta = 0$ and $\eta = 0.3$, G_{hh}/G_{hv} decreases during loading and eventually approaches 1. When $\eta = 0.6$, G_{hh}/G_{hv} remains nearly constant, indicating that the initial fabric anisotropy is close to the critical state fabric anisotropy for this stress ratio. These variations confirm that the degree of fabric anisotropy at the critical state depends on the stress ratio, as shown in Fig. 4(c).

4.4. Simulations of strength anisotropy under true triaxial loadings

A series of true triaxial tests reported in the literature is also simulated using the generalized strength criterion to validate the model. Under true triaxial loading conditions, the material fabric may evolve towards orthotropy. Therefore, an orthotropic fabric tensor $\mathbf{F} = \text{diag}(F_{11}, F_{22}, F_{33})$ is adopted in simulations. Additionally, the Mises and SMP criteria are considered in the generalized strength criterion. The model parameters used in simulations are listed in Table 2.

Fig. 5 compares simulation and experimental results for strength anisotropy under true triaxial loading. The figure shows that the anisotropic elastoplastic model combined with the generalized strength criterion generally agrees well with the experimental results. The effect of fabric anisotropy on strength anisotropy differs for different soils. For the San Francisco Bay mud and glass beads, significant strength anisotropy is observed because the strength in sectors II and III is much lower than that in sector I. In contrast, the Cambria sand and Santa Monica Beach sand exhibit relatively minor strength anisotropy.

In comparison, simulations with the anisotropic SMP criterion are also shown in Fig. 5. It can be observed that the anisotropic SMP criterion underestimates the strength anisotropy for the mid-range of intermediate stress ratios, particularly for the Cambria sand and glass beads. This underestimation highlights the need to combine multiple strength criteria to accurately capture the effect of intermediate principal stress on the strength anisotropy of granular materials.

4.5. Simulations of drained triaxial tests

Fig. 6 shows comparisons between simulations and experimental results for drained triaxial tests on Cambria sand under different loading directions. The anisotropic elastoplastic model accurately captures the effect of fabric anisotropy on the strength of Cambria sand under different loading directions. Because the Cambria sand is initially transversely isotropic, the strength under coaxial loading ($\alpha = 0^\circ$) is higher than that under non-coaxial loading ($\alpha = 90^\circ$). Therefore, it can be observed in Fig. 6(a) and (b) that the deviatoric stress and volumetric strain under coaxial loading are higher than those under non-coaxial loading.

The fabric anisotropy variable A shown in Fig. 6(c) indicates that $A > 0$ for coaxial loading and $A < 0$ for non-coaxial loading initially, suggesting that the fabric tensor is aligned with the loading direction for coaxial loading and is opposite to the loading direction for non-coaxial loading. As shearing continues, A approaches 1 in both loading directions, indicating that the fabric tensor aligns with the loading direction. This suggests that the material fabric tends to rotate to align with the loading direction, enabling the granular assembly to resist the applied load. Thanks to the incorporation of the fabric evolution law as shown in Eq. (19), the anisotropic elastoplastic model can inherently capture the effect of the fabric anisotropy on the macroscopic response of granular materials. The norm of the fabric tensor F , as shown in Fig. 6(d), increases towards 1 for coaxial loading, whereas it first decreases slightly and then increases towards 1 for non-coaxial loading. This occurs because a non-coaxial loading direction relative to the material fabric induces a structural reorganization that increases the number of particles aligned with the external load while decreasing the number of particles in the perpendicular direction. As a result, the material tends to be isotropic in the initial stage of loading, and then fabric anisotropy develops as shearing continues.

To this end, it can be concluded that the proposed anisotropic elastoplastic framework can be applied to generalize existing elastoplastic models to account for the effects of fabric anisotropy and intermediate principal stress on the mechanical behavior of granular materials by using the anisotropic compliance tensor, the transformed stress tensor, and the transformed friction angle.

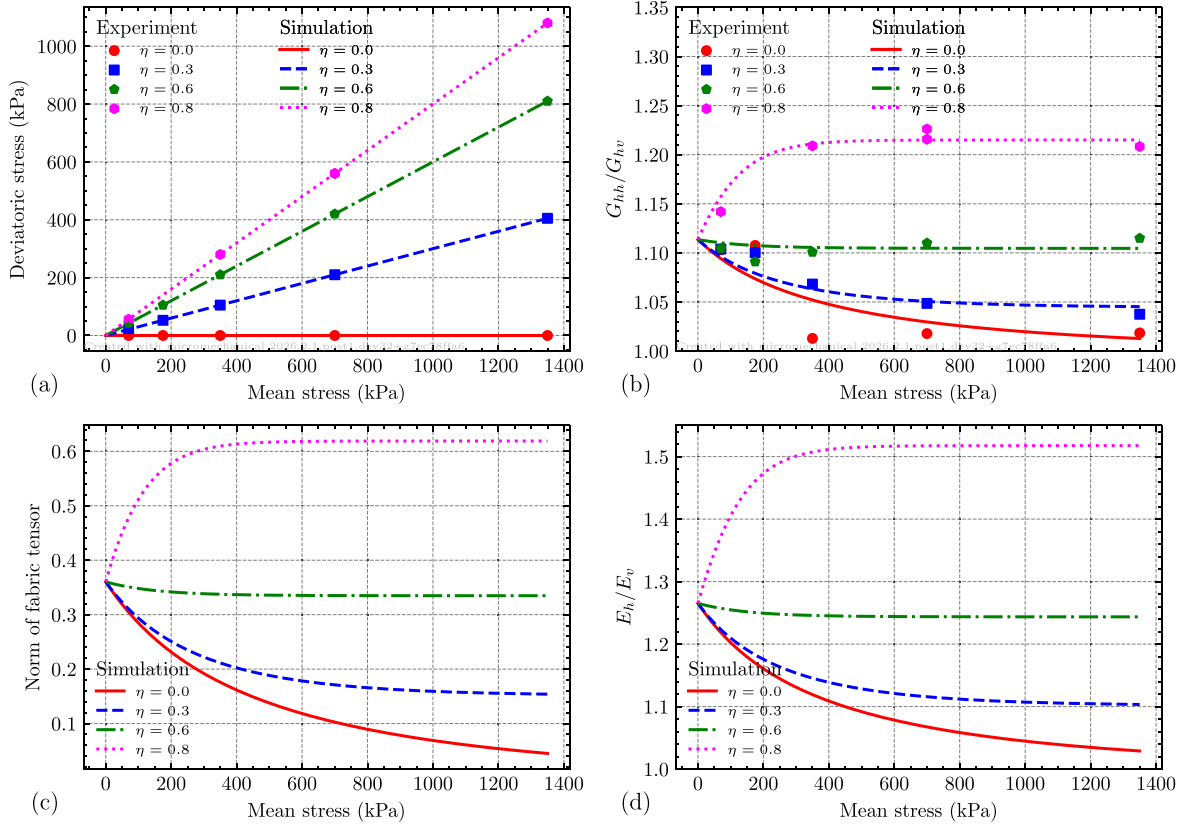


Fig. 4. Comparison between simulations and experimental results (Mitaritonna et al., 2014) for anisotropic elastic properties of Lucera clay: (a) stress path; (b) shear stiffness ratio; (c) norm of fabric tensor; (d) Young's modulus ratio.

5. Extension to critical-state-based elastoplastic models

While the introduction of the transformed stress tensor in the yield function and plastic potential function enables the model to inherently capture the effects of fabric anisotropy on the elastic and strength behaviors of granular materials, it may lead to violations of the anisotropic critical state theory (ACST) (Li and Dafalias, 2012) when applied to critical-state-based elastoplastic models that consider the critical state line. Therefore, this section proposes an additional modification to the anisotropic elastoplastic framework to ensure compatibility with the ACST framework when applied to critical-state-based elastoplastic models.

5.1. Origin of the issue

According to the plastic potential function defined in Eq. (25), the plastic strain increment can be expressed as:

$$d\epsilon^p = d\lambda \frac{\partial g}{\partial \sigma_{kl}^t} \frac{\partial \sigma_{kl}^t}{\partial \sigma_{ij}} = d\lambda \left(\frac{\partial g}{\partial p^t} \frac{\partial p^t}{\partial \sigma_{kl}^t} \frac{\partial \sigma_{kl}^t}{\partial \sigma_{ij}} + \frac{\partial g}{\partial q^t} \frac{\partial q^t}{\partial \sigma_{kl}^t} \frac{\partial \sigma_{kl}^t}{\partial \sigma_{ij}} \right) \quad (32)$$

where $s^t = \sigma^t - p^t \mathbf{I}$ is the deviatoric part of the transformed stress tensor. The derivative of p^t and q^t with respect to σ_{kl}^t can be expressed as:

$$\begin{aligned} \frac{\partial p^t}{\partial \sigma_{kl}^t} &= \frac{1}{3} \delta_{kl} \\ \frac{\partial q^t}{\partial \sigma_{kl}^t} &= \begin{bmatrix} \frac{2-b^t}{2\sqrt{(b^t)^2 - b^t + 1}} & 0 & 0 \\ 0 & \frac{2b^t - 1}{2\sqrt{(b^t)^2 - b^t + 1}} & 0 \\ 0 & 0 & \frac{-1 - b^t}{2\sqrt{(b^t)^2 - b^t + 1}} \end{bmatrix} \\ &= \sqrt{\frac{3}{2}} \frac{s^t}{\|s^t\|} = \sqrt{\frac{3}{2}} \mathbf{n} \end{aligned} \quad (33)$$

where δ_{kl} is the Kronecker delta. Substituting Eq. (33) into Eq. (32), the plastic volumetric strain increment can be derived as:

$$\begin{aligned} d\epsilon_v^p &= d\lambda \cdot A_d \left(M_{pt}^t - \frac{q^t}{p^t} \right) \frac{\text{tr}(\mathbf{A}^n)}{3} - A_q^n \\ A_q^n &= \sqrt{\frac{3}{2}} \mathbf{A}^n : \mathbf{n} = \frac{A_{22}^n(1 - 2b^t) - A_{11}^n(2 - b^t) + A_{33}^n(b^t + 1)}{2\sqrt{(b^t)^2 - b^t + 1}} \sqrt{\frac{3}{2}} \end{aligned} \quad (34)$$

where $\mathbf{A}^n$ is a fabric tensor defined in Eq. (6).

Eq. (34) shows that the effect of fabric anisotropy on the plastic volumetric strain is accounted for through the fabric tensor $\mathbf{A}^n$. Under isotropic conditions, Eq. (34) reduces to the plastic volumetric strain of the isotropic elastoplastic model defined in Eq. (27). Under anisotropic conditions, the fabric tensor evolves to its critical-state value at the anisotropic critical state according to the ACST framework, and the dilatancy state line reduces to the critical-state line at the critical state. Eq. (34) shows that a more dilative response is introduced through the term A_q^n in the stress-dilatancy relationship under anisotropic conditions, which conforms with experimental observations (Ochiai and Lade, 1983). However, it does not converge to the critical state because of the term A_q^n , leading to continuous changes of the volumetric strain and void ratio at the critical state, which violates the ACST framework. Moreover, the term A_q^n increases with the degree of fabric anisotropy, indicating that the violation of ACST becomes more significant during shearing when the fabric anisotropy evolves to the critical state. It is interesting to note that this phenomenon is also observed in the micromechanical model that inspired the anisotropic elastoplastic framework proposed in this paper, where the macroscopic stress-dilatancy relationship derived from the microscopic force-dilatancy relationship does not converge to the critical state. Wang et al. (2025) introduced a correction to the strain-averaging formulation of the micromechanical model to ensure convergence of the stress-dilatancy relationship to the critical state. However, this approach violates the principle of energy conservation, and an additional

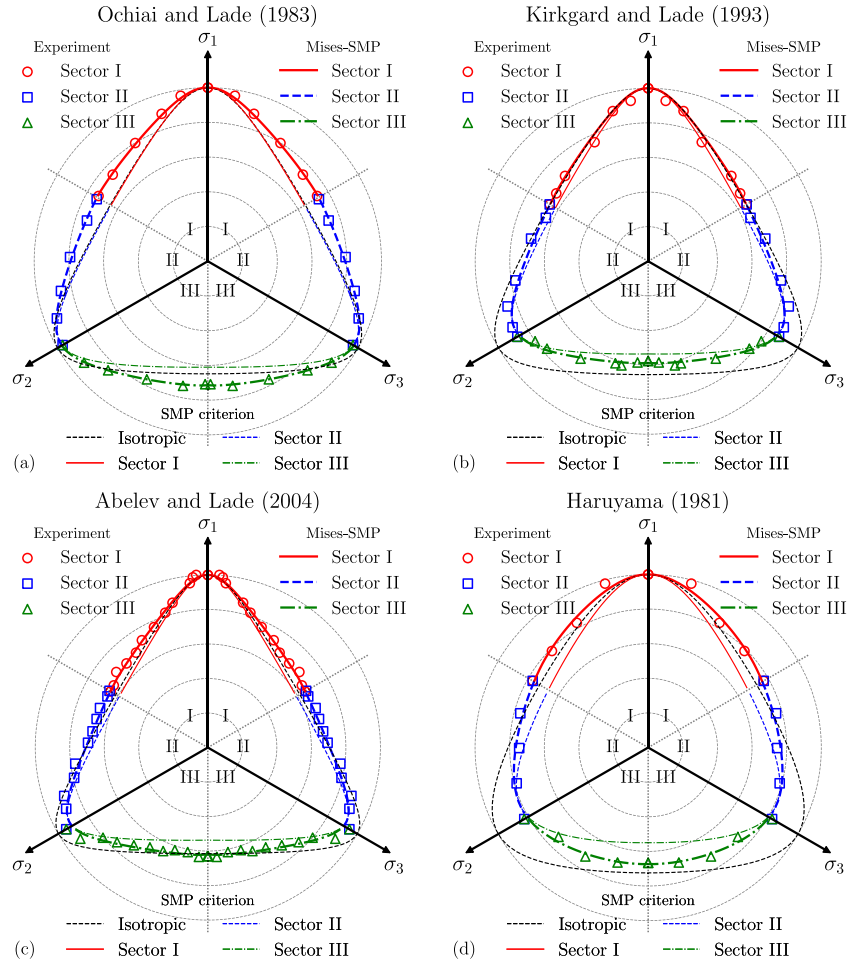


Fig. 5. Comparison between simulations and experimental results for strength anisotropy of true triaxial tests: (a) Ochiai and Lade (1983); (b) Kirkgard and Lade (1993); (c) Abelev and Lade (2004); (d) Haruyama (1981).

iterative algorithm is required to enforce energy conservation during the correction process.

5.2. Revised anisotropic elastoplastic framework

As shown in Fig. 7, the ACST framework introduces an additional dilatancy state line that converges to the critical state line at the anisotropic critical state to consider the effect of fabric anisotropy on the dilatancy behavior of granular materials. The anisotropy parameter (AP), referred as ψ_A , is defined as a multiplicative decomposition of the dependence of $\hat{\psi}_A(e, p, A)$ on e , p , and A :

$$\psi_A = \hat{\psi}_A(e, p, A) = \hat{e}_A(e, p) \hat{\psi}_A(A) = \hat{e}_A(e, p)(A - 1) \quad (35)$$

where $\hat{e}_A(e, p)$ and $\hat{\psi}_A(A)$ are functions of e , p , and A , respectively. In Li and Dafalias (2012), it is assumed that $\hat{e}_A(e, p) = e_A$, and the dilatancy state line is observed as a parallel translation of the critical state line in the e - p space, as shown in Fig. 7. The dilatancy state parameter (DSP), $\zeta = e - e_d = \psi - \psi_A$, is introduced to measure the distance between the current state and the dilatancy state line.

The ACST framework provides a general method for considering the effect of fabric anisotropy on the mechanical behavior of granular materials by introducing the new dilatancy state parameter ζ into critical-state-based elastoplastic models. However, the effect of fabric anisotropy is only considered in the critical state-related behavior of granular materials, and the effect of fabric anisotropy on the elastic and strength behaviors is not explicitly considered in the ACST framework. As discussed in previous sections, the anisotropic

elastoplastic framework proposed in this paper can inherently capture the effect of fabric anisotropy on the elastic and strength behaviors of granular materials. Moreover, the term A_q^n in the stress-dilatancy relationship under anisotropic conditions introduces an additional plastic volumetric strain, leading to additional changes in void ratio during shearing. For dense sand or overconsolidated clay, the state parameter ψ decreases during shearing. Although the proposed anisotropic elastoplastic framework does not explicitly define the dilatancy state line, the state parameter ψ can be treated as the dilatancy state parameter ζ when the increase in void ratio due to the term A_q^n equals the anisotropy parameter ψ_A . According to Eq. (34), the increase in the state parameter ψ due to the term A_q^n can be expressed as:

$$d\psi = de = -\sqrt{\frac{3}{2}}(1 + e_0)A^n : \mathbf{n} = -\sqrt{\frac{3}{2}}(1 + e_0) \left(\mathbf{I} + \frac{2}{5}\mathbf{F} \right)^{-1} : \mathbf{n} \quad (36)$$

where e_0 is the initial void ratio. The above equation can be simplified by the Neumann series expansion when the degree of fabric anisotropy is small, leading to:

$$d\psi \approx -\sqrt{\frac{3}{2}}(1 + e_0) \left(\mathbf{I} - \frac{2}{5}\mathbf{F} \right) : \mathbf{n} = \frac{2}{5}\sqrt{\frac{3}{2}}(1 + e_0)\mathbf{F} : \mathbf{n} = k\mathbf{F} : \mathbf{n} \quad (37)$$

Eq. (37) shows that the increase in void ratio is proportional to the deviatoric fabric tensor $\mathbf{F}$, leading to substantial changes in the void ratio during shearing when the fabric anisotropy evolves to the critical state. Li and Dafalias (2012) defines the anisotropy parameter ψ_A as a linear function of $A - 1$. To be consistent with the ACST framework, the following fabric-load coupling tensor is proposed:

$$\mathbf{C} = \mathbf{F} - \mathbf{n} \quad (38)$$

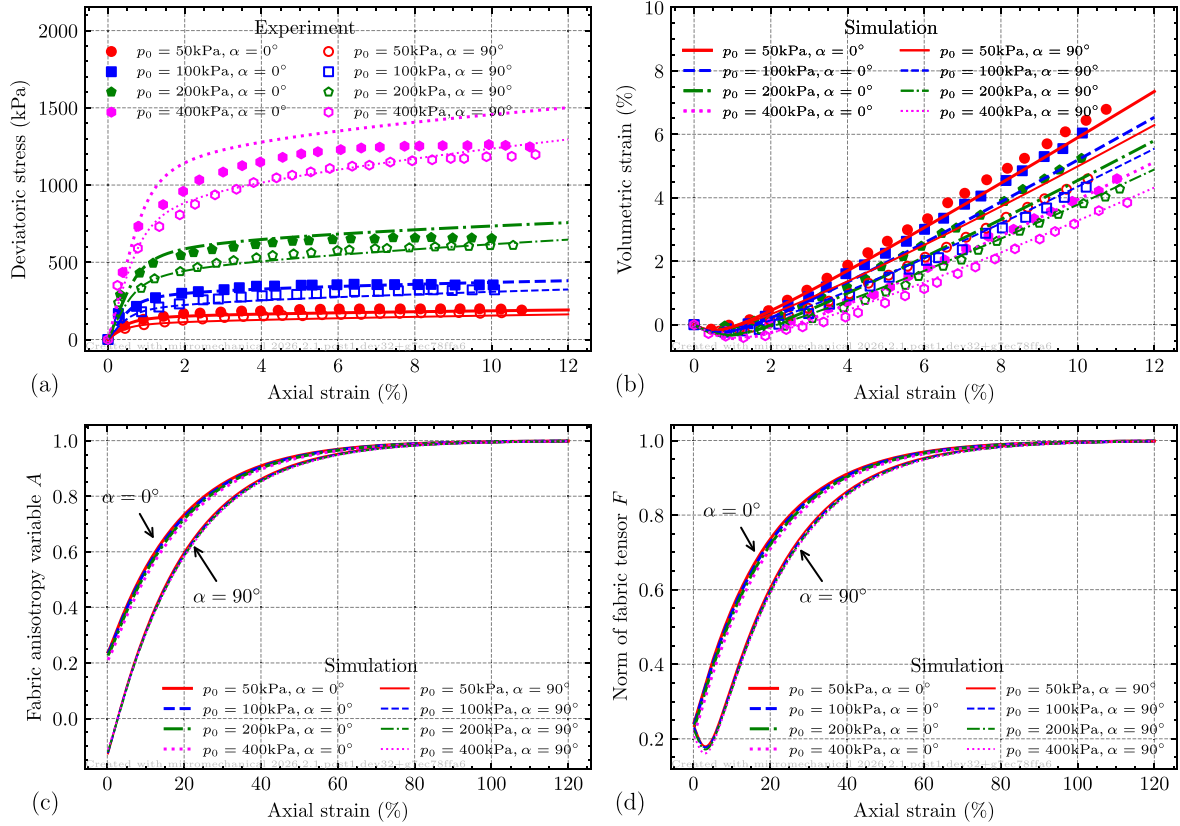


Fig. 6. Comparison between simulations and experimental results (Ochiai and Lade, 1983) for drained triaxial tests on Cambria sand: (a) deviatoric stress; (b) volumetric strain; (c) fabric anisotropy variable; (d) norm of fabric tensor.

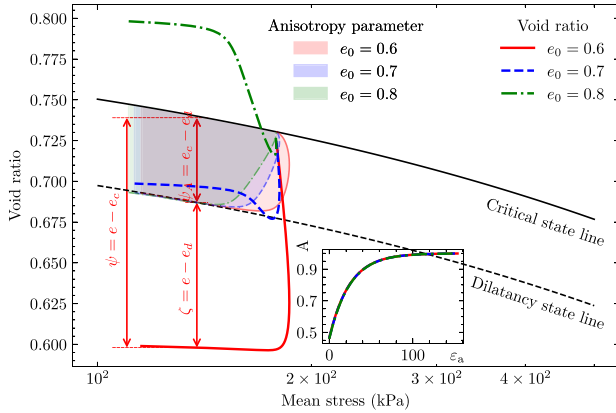


Fig. 7. Schematic diagram of the anisotropic critical state theory.

The fabric-load coupling tensor $\mathbf{C}$ is defined as the difference between the fabric tensor $\mathbf{F}$ and the loading direction $\mathbf{n}$, which measures the potential of the structural reorganization towards the loading direction. Based on the fabric-load coupling tensor $\mathbf{C}$, the increase in void ratio can be expressed as:

$$de = k\mathbf{C} : \mathbf{n} = k(\mathbf{F} : \mathbf{n} - 1) = k(A - 1) \quad (39)$$

Eq. (39) shows that the increase in void ratio is a linear function of $A - 1$, which is consistent with the definition of the anisotropy parameter ψ_A in the ACST framework. It can be observed that although Li and Dafalias (2012) defined the anisotropy parameter ψ_A with the scalar fabric anisotropy variable A , it is equivalent to the proposed

approach of defining the fabric-load coupling tensor $\mathbf{C}$ that measures the difference between the fabric tensor $\mathbf{F}$ and the loading direction $\mathbf{n}$. The fabric-load coupling tensor $\mathbf{C}$ is then used to modify the stress tensor $\boldsymbol{\sigma}$ to obtain a new combined stress tensor $\boldsymbol{\sigma}^c$ as:

$$\boldsymbol{\sigma}^c = \left\{ \boldsymbol{\sigma} \left(\mathbf{I} + \frac{2}{5} \mathbf{C} \right)^{-1} \right\}_{\text{sym}} = \left\{ \boldsymbol{\sigma} \left[\mathbf{I} + \frac{2}{5} (\mathbf{F} - \mathbf{n}) \right]^{-1} \right\}_{\text{sym}} \quad (40)$$

where the subscript “sym” denotes the symmetric part of the tensor. The combined stress tensor $\boldsymbol{\sigma}^c$ is a function of the original stress tensor $\boldsymbol{\sigma}$, the fabric tensor $\mathbf{F}$, and the loading direction $\mathbf{n}$. Based on this equation, two boundaries can be found:

$$\boldsymbol{\sigma}^c = h(\boldsymbol{\sigma}, \mathbf{F}, \mathbf{n}) = \begin{cases} \boldsymbol{\sigma}^i, & \mathbf{n} = \mathbf{0} \text{ (At isotropic stress state)} \\ \boldsymbol{\sigma}, & \mathbf{F} = \mathbf{n} \text{ (At critical state)} \end{cases} \quad (41)$$

Eq. (41) shows that the combined stress tensor $\boldsymbol{\sigma}^c$ reduces to the transformed stress tensor $\boldsymbol{\sigma}^i$ at isotropic stress state ($\mathbf{n} = \mathbf{0}$), and it reduces to the original stress tensor $\boldsymbol{\sigma}$ as the fabric anisotropy evolves to the critical state ($\mathbf{F} = \mathbf{n}$), ensuring that the effect of fabric anisotropy on the dilatancy behavior of granular materials vanishes at the critical state. Fig. 8 shows the schematic diagram of revised anisotropic elastoplastic framework.

The implementation of the revised anisotropic elastoplastic framework is shown in Fig. 9. The anisotropic elasticity and anisotropic strength are captured by the transformed stress tensor $\boldsymbol{\sigma}^i$ in the anisotropic elastic properties and the yield function, respectively. The effect of fabric anisotropy on the dilatancy behavior is captured by the fabric-load coupling tensor $\mathbf{C}$ and implemented through the combined stress tensor $\boldsymbol{\sigma}^c$ in the flow rule.

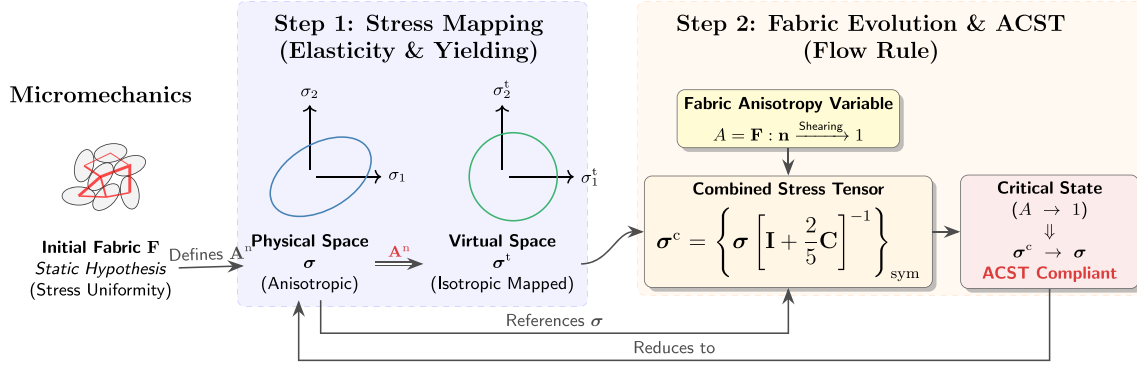


Fig. 8. Schematic diagram of revised anisotropic elastoplastic framework.

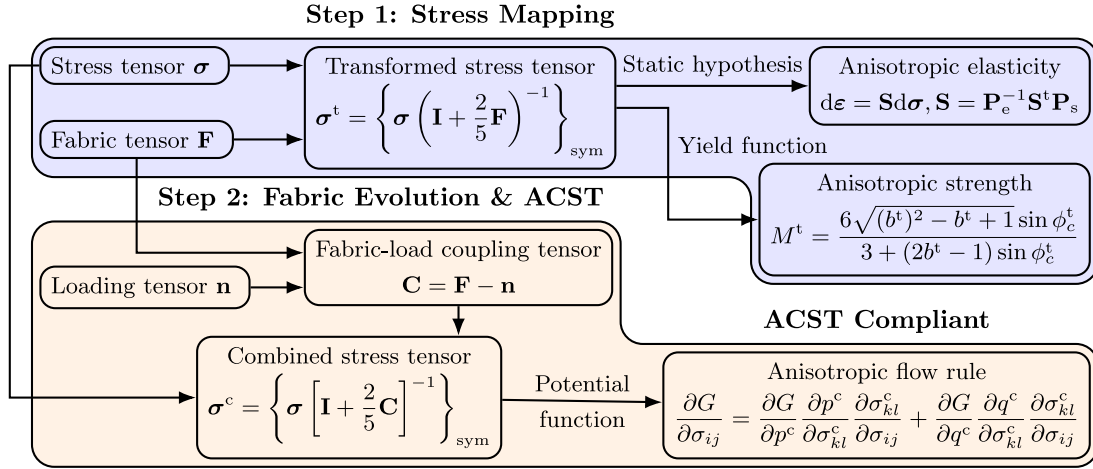


Fig. 9. Implementation of revised anisotropic elastoplastic framework.

5.3. Consideration of the non-proportionality between fabric tensor and loading direction

Li and Dafalias (2020) and Dafalias and Li (2026) stated that the fabric anisotropy variable A in Eq. (20) cannot consider the non-proportional parts of the fabric tensor $\mathbf{F}$ in regard to the loading direction $\mathbf{n}$. As shown in Fig. 10, the fabric tensor $\mathbf{F}$ can be decomposed into the following three parts:

$$\begin{aligned} \mathbf{F} &= \mathbf{F}_{pr} + \mathbf{F}_{or} + \mathbf{F}_{nc} = (\mathbf{F} : \mathbf{n})\mathbf{n} + (\mathbf{F} : \mathbf{m})\mathbf{m} + (\mathbf{F} : \mathbf{z})\mathbf{z} \\ \mathbf{m} &= \frac{\mathbf{n}^2 - (\text{tr} \mathbf{n}^3)\mathbf{n} - \mathbf{I}/3}{\sqrt{1/6 - \text{tr}^2 \mathbf{n}^3}}, \mathbf{z} = \frac{\mathbf{F} - (\mathbf{F} : \mathbf{n})\mathbf{n} - (\mathbf{F} : \mathbf{m})\mathbf{m}}{\sqrt{\mathbf{F} : \mathbf{F} - (\mathbf{F} : \mathbf{n})^2 - (\mathbf{F} : \mathbf{m})^2}} \end{aligned} \quad (42)$$

where $\mathbf{F}_{pr}$, $\mathbf{F}_{or}$, and $\mathbf{F}_{nc}$ are the proportional, orthogonal, and non-coaxial parts of the fabric tensor $\mathbf{F}$ in regard to the loading direction $\mathbf{n}$, respectively; $\mathbf{m}$ and $\mathbf{z}$ are the unit direction tensors of $\mathbf{F}_{or}$ and $\mathbf{F}_{nc}$, respectively.

The fabric anisotropy variable A in Eq. (20) only considers the proportional part $\mathbf{F}_{pr}$ of the fabric tensor $\mathbf{F}$ in regard to the loading direction $\mathbf{n}$, which may lead to an underestimation of the effect of fabric anisotropy on the dilatancy behavior of granular materials. Dafalias and Li (2026) proposed a modified fabric anisotropy variable to consider the non-proportional parts of the fabric tensor $\mathbf{F}$ in regard to the loading direction $\mathbf{n}$:

$$A = \mathbf{F} : \mathbf{n} + a_{or} \mathbf{F} : \mathbf{m} + a_{nc} \mathbf{F} : \mathbf{z} \quad (43)$$

where a_{or} and a_{nc} are parameters to control the effect of the orthogonal and non-coaxial parts of the fabric tensor $\mathbf{F}$ in regard to the loading direction $\mathbf{n}$, respectively. Accordingly, the fabric-load coupling tensor $\mathbf{C}$ in Eq. (38) can be modified as:

$$\mathbf{C} = \mathbf{F} + a_{or}(\mathbf{F} : \mathbf{m})\mathbf{n} + a_{nc}(\mathbf{F} : \mathbf{z})\mathbf{n} - \mathbf{n} \quad (44)$$

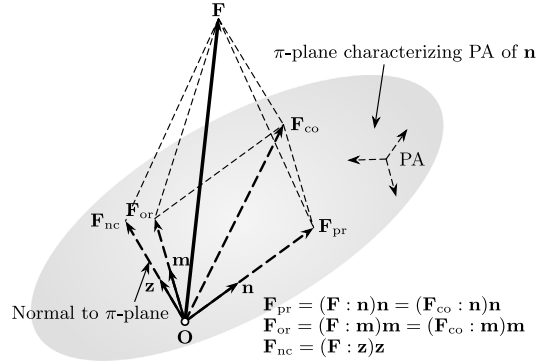


Fig. 10. Decomposition of the fabric tensor in regard to the loading direction. Source: After Li and Dafalias (2020).

At critical state, it follows that $\mathbf{F} = \mathbf{n}$, $\mathbf{F} : \mathbf{m} = 0$, and $\mathbf{F} : \mathbf{z} = 0$, leading to $\mathbf{C} = \mathbf{0}$, which ensures that the effect of fabric anisotropy on the dilatancy behavior of granular materials vanishes at the critical state.

5.4. Anisotropic SIMSAND model

Based on the revised anisotropic elastoplastic framework, an anisotropic SIMSAND model is proposed. The yield function and stress-dilatancy relationship of the anisotropic SIMSAND model are the same as those defined in Eqs. (24) and (25), except that the stress tensor is replaced by the stress tensor σ^c in the stress-dilatancy relationship.

Table 3
Formulations of the SIMSAND model and the anisotropic SIMSAND model.

Category	SIMSAND model	Anisotropic SIMSAND model
Elasticity	$d\epsilon_v^e = dp/K, d\epsilon_v^e = dq/(3G)$ $G = G_0 \frac{(2.97 - e)^2}{1 + e} \left(\frac{p}{p_a} \right)^n$	$d\epsilon = S d\sigma, S = P_e^{-1} S' P_s$ $G = G_0 \frac{(2.97 - e)^2}{1 + e} \left(\frac{p^t}{p_a} \right)^n$
Critical state	$e_c = e_{ref} - \lambda (p/p_a)^\xi$	$e_c = e_{ref} - \lambda (p/p_a)^\xi$
Stress-dilatancy	$D = A_d \left(M_{pt} - \frac{q}{p} \right)$	$D = A_d \left(M_{pt}^t - \frac{q^t}{p^t} \right)$
Density effect	$M_p = (e_c/e)^{n_p} M, M_{pt} = (e/e_c)^{n_d} M$ $M = \frac{6\sqrt{b^2 - b + 1} \sin \phi_c}{3 + (2b - 1) \sin \phi_c}$	$M_p^t = (e_c/e)^{n_p} M^t, M_{pt}^t = (e/e_c)^{n_d} M^t$ $M^t = \frac{6\sqrt{(b^t)^2 - b^t + 1} \sin \phi_c^t}{3 + (2b^t - 1) \sin \phi_c^t}$
Yield function	$F(\sigma) = \frac{q}{p} - \frac{M_p \epsilon_d^p}{k_p + \epsilon_d^p}$ $\frac{\partial f}{\partial \sigma_{ij}} = \frac{\partial f}{\partial p} \frac{\partial p}{\partial \sigma_{ij}} + \frac{\partial f}{\partial q} \frac{\partial q}{\partial \sigma_{ij}}$ $\frac{\partial f}{\partial p} = -\frac{q}{p}, \frac{\partial f}{\partial q} = \frac{1}{p}$	$F(\sigma^t) = \frac{q^t}{p^t} - \frac{M_p^t \epsilon_d^p}{k_p + \epsilon_d^p}$ $\frac{\partial f}{\partial \sigma_{ij}} = \frac{\partial f}{\partial p^t} \frac{\partial p^t}{\partial \sigma_{ij}} + \frac{\partial f}{\partial q^t} \frac{\partial q^t}{\partial \sigma_{ij}}$ $\frac{\partial f}{\partial p^t} = \frac{\partial f}{\partial p} (p^t, q^t), \frac{\partial f}{\partial q^t} = \frac{\partial f}{\partial q} (p^t, q^t)$
Potential function	$\frac{\partial g}{\partial \sigma_{ij}} = \frac{\partial g}{\partial p} \frac{\partial p}{\partial \sigma_{ij}} + \frac{\partial g}{\partial q} \frac{\partial q}{\partial \sigma_{ij}}$ $\frac{\partial g}{\partial p} = D, \frac{\partial g}{\partial q} = 1$	$\frac{\partial g}{\partial \sigma_{ij}} = \frac{\partial g}{\partial p^t} \frac{\partial p^t}{\partial \sigma_{ij}} + \frac{\partial g}{\partial q^t} \frac{\partial q^t}{\partial \sigma_{ij}}$ $\frac{\partial g}{\partial p^t} = D, \frac{\partial g}{\partial q^t} = 1$

Table 4
Summary of model parameters for anisotropic SIMSAND model (unit: G_0 in MPa, ϕ_c in degrees).

Soil type	G_0	ν	n	ϕ_c	k_p	e_{ref}	λ	ξ	n_p	n_d	A_d	Δ	k_f
Toyoura sand	7.4	0.2	0.5	26	4e-4	0.934	0.019	0.7	0.87	1.88	2.19	0.4	4

To consider the density effect of soils, the peak stress ratio M_p^t , and phase-transformation stress ratio M_{pt}^t are functions of the void ratio e :

$$M_p^t = \left(\frac{e_c}{e} \right)^{n_p} M^t, M_{pt}^t = \left(\frac{e}{e_c} \right)^{n_d} M^t \quad (45)$$

where n_p and n_d are parameters to control the density effect on the peak stress ratio and phase-transformation stress ratio, respectively. The critical state line is given as:

$$e_c = e_{ref} - \lambda (p/p_a)^\xi \quad (46)$$

where e_{ref} is the reference void ratio, λ and ξ are parameters to control the shape of the critical state line. Table 3 shows formulations of the SIMSAND model and the anisotropic SIMSAND model.

6. Model verifications

The anisotropic SIMSAND model requires several parameters to be calibrated using experimental data. Most parameters can be calibrated using the above calibration procedures for the simple anisotropic elastoplastic model. Additional parameters for the critical state line (e_{ref} , λ , and ξ) can be calibrated from the drained triaxial test results at different confining pressures. In the following sections, a series of experiments by Yoshimine et al. (1998) is simulated to verify the performance of the anisotropic SIMSAND model. The model parameters shown in Table 4 are calibrated using the procedures above.

6.1. Simulations of torsional shear tests

Fig. 11 compares simulations based on the SMP criterion with the experimental stress-strain relationships for torsional shear tests on Toyoura sand, and Fig. 12 shows the evolution of the fabric anisotropy variable and norm of the fabric tensor at very large strain for the same tests. Overall, the anisotropic SIMSAND model can effectively capture the effect of fabric anisotropy on the mechanical response of granular materials under various loading paths, regardless of the intermediate principal stress ratio or loading direction. For triaxial compression ($b = 0$), triaxial extension ($b = 1$), and true triaxial conditions with

$b = 0.5$, the Toyoura sand exhibits higher strength under coaxial loading compared to non-coaxial loading. As α increases from 0° to 90° , the mechanical response of the Toyoura sand becomes softer and more contractive. The evolution of the fabric anisotropy variable A and the norm of the fabric tensor F also shows trends similar to those of the Cambria sand. For the case of $\alpha = 90^\circ$, the norm of the fabric tensor F decreases significantly to zero before increasing again. The intermediate principal stress ratio b also significantly affects the mechanical response of Toyoura sand, as shown in Fig. 11(c) and (f). For true triaxial conditions with higher b values, the sand exhibits softer, more contractive behavior than that with lower b values. Although the loading direction α is fixed at 45° , the intermediate principal stress increases along with the increase of b , which leads to larger external loading in the direction of the fabric's minor principal direction. This results in a more contractive response, as shown in Fig. 11(c) and (f). As shown in Fig. 12(c) and (f), A is generally smaller under true triaxial conditions with higher b values than that with lower b values, indicating that the fabric is less aligned with the loading direction.

Fig. 13 shows the evolution of normalized deviatoric stress at very large strains for torsional shear tests on Toyoura sand under different loading paths. As shown in Fig. 13, the deviatoric stress normalized by the critical stress ratio converges to the same value at the critical state for all loading paths, regardless of the intermediate principal stress ratio or loading direction. This indicates that the critical state is independent of the loading path and of inherent fabric anisotropy, consistent with the ACST framework. However, the evolution of the normalized deviatoric stress towards the critical state is highly dependent on the loading path and fabric anisotropy. In general, the normalized deviatoric stress converges to the critical state faster at lower b and smaller α . This is because loading paths with lower b and smaller α are more aligned with the fabric's major principal direction, leading to faster evolution of the fabric anisotropy and faster convergence to the critical state.

Fig. 14 shows the evolution of fabric anisotropy at very large strain for torsional shear tests on Toyoura sand under different loading paths. The loading direction tensor and fabric tensor are visualized in 3D space to illustrate the evolution of fabric anisotropy during shearing. Initially, the fabric is misaligned with the loading direction, depending on the intermediate principal stress ratio and loading direction. As

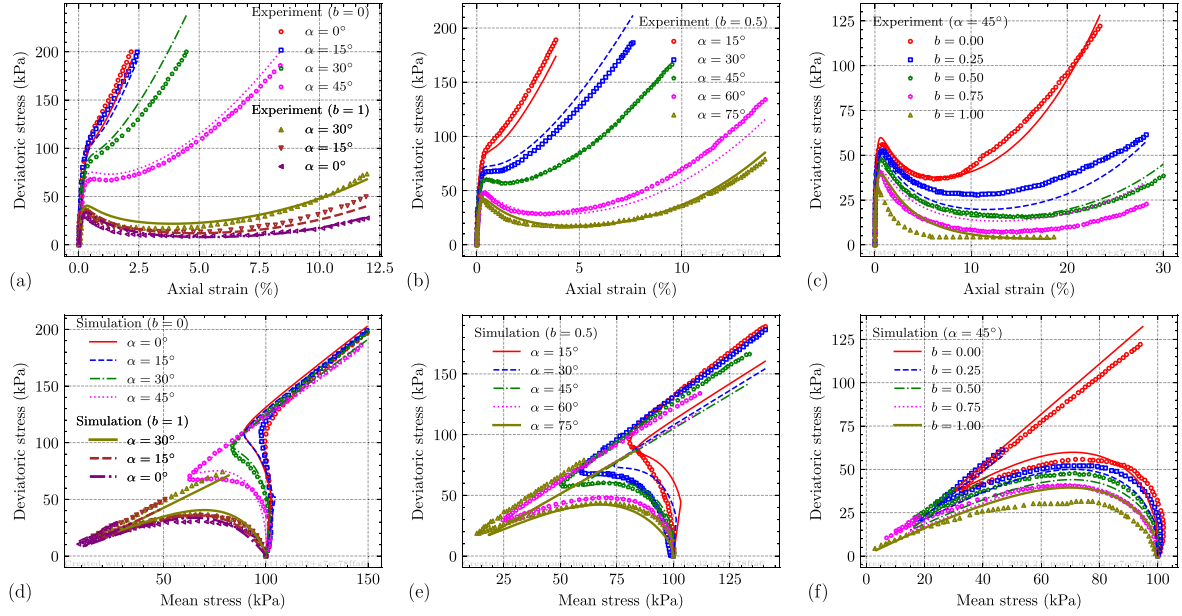


Fig. 11. Comparison between simulations and experimental results (Yoshimine et al., 1998) for torsional shear tests on Toyoura sand: (a)–(c) stress–strain relationship; (d)–(f) stress path.

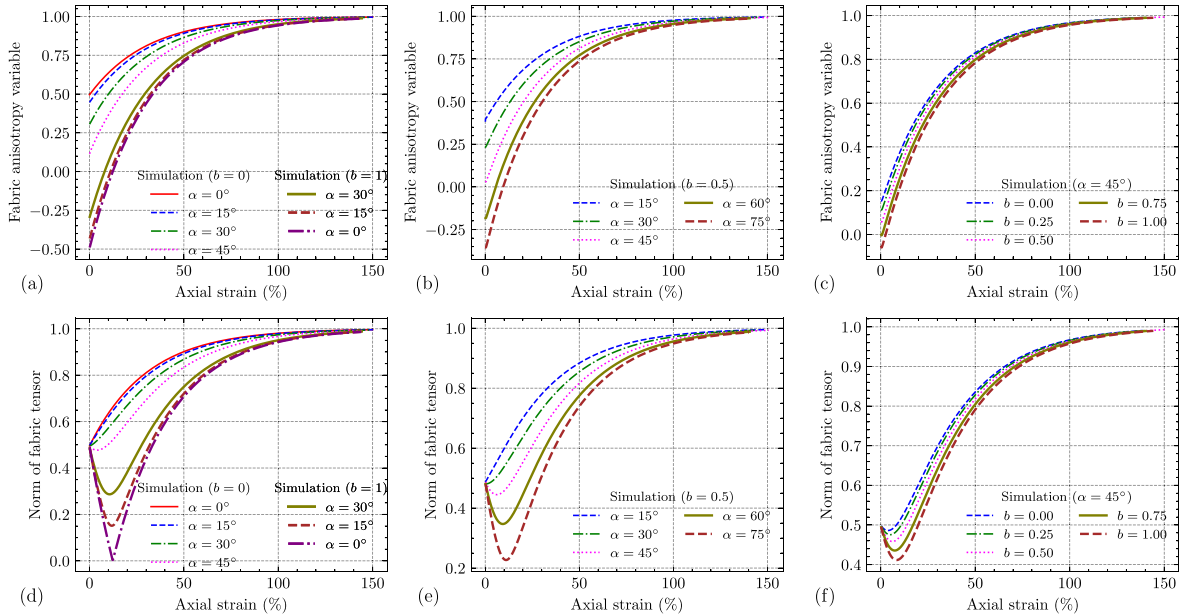


Fig. 12. Evolution of fabric anisotropy variable and norm of fabric tensor at very large strain for torsional shear tests on Toyoura sand: (a)–(c) fabric anisotropy variable; (d)–(f) norm of fabric tensor.

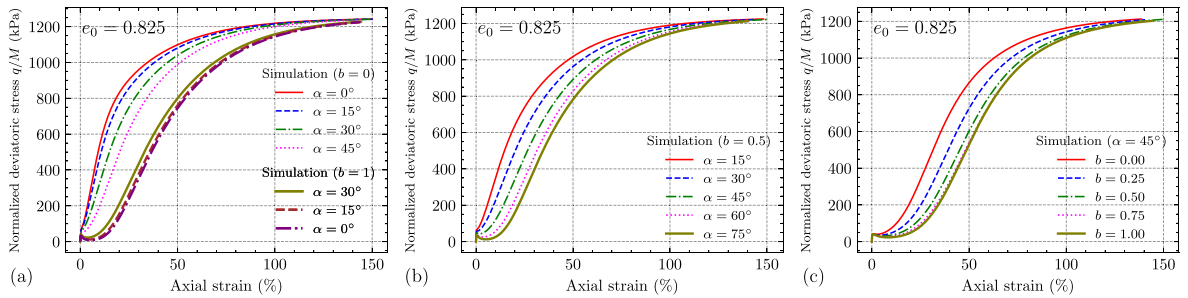


Fig. 13. Evolution of normalized deviatoric stress at very large strains for torsional shear tests on Toyoura sand: (a) triaxial compression and extension loadings with different loading directions; (b) true triaxial loadings with $b = 0.5$ and different loading directions; (c) true triaxial loadings with $\alpha = 45^\circ$ and different b values.

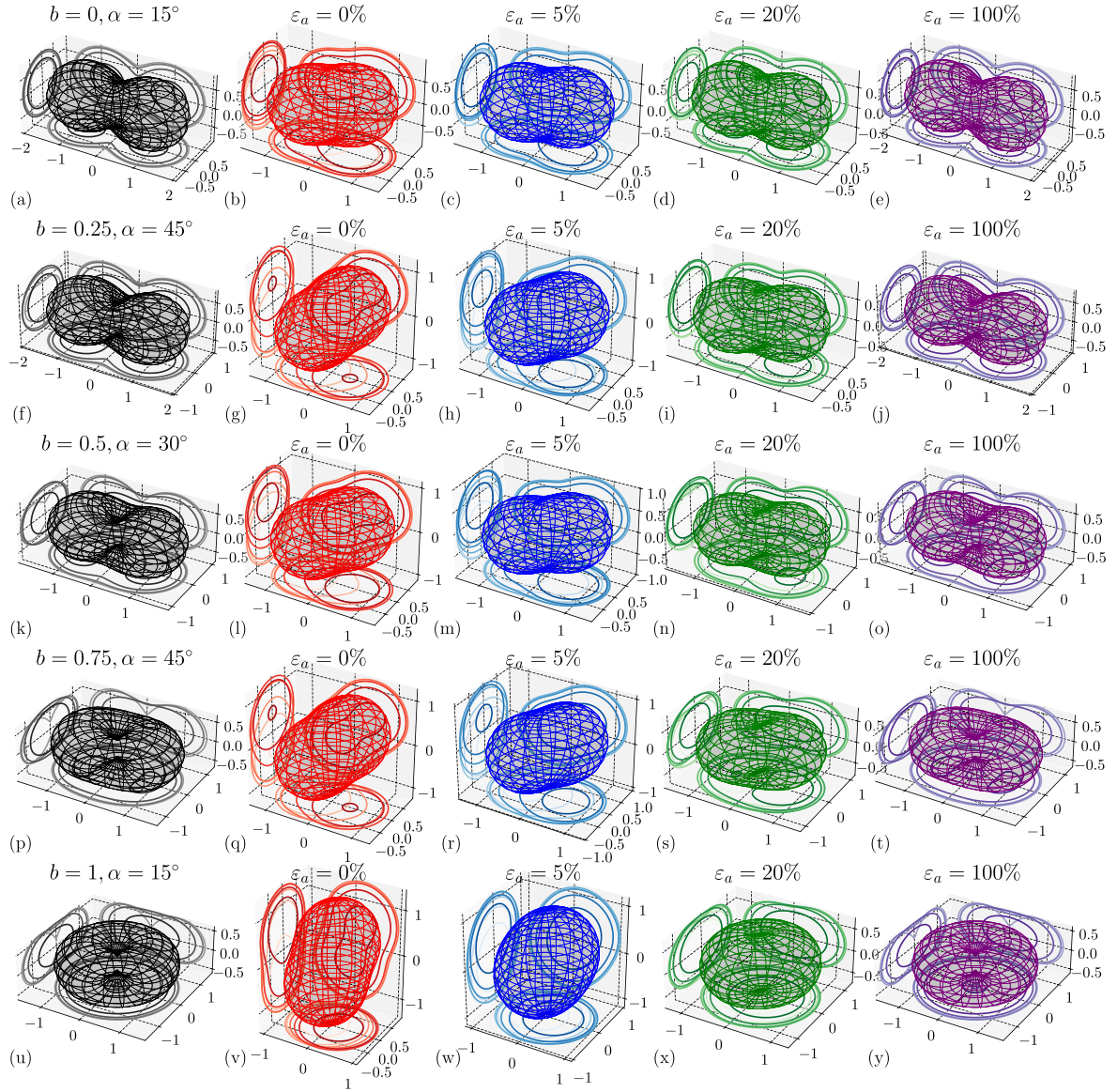


Fig. 14. Evolution of loading direction tensor and fabric tensor with different axial strains at very large strain for torsional shear tests on Toyoura sand: (a)–(e) $b = 0, \alpha = 15^\circ$; (f)–(j) $b = 0.25, \alpha = 45^\circ$; (k)–(o) $b = 0.5, \alpha = 30^\circ$; (p)–(t) $b = 0.75, \alpha = 45^\circ$; (u)–(y) $b = 1, \alpha = 75^\circ$.

shearing proceeds, the fabric eventually becomes fully aligned with the loading direction at the critical state, regardless of the intermediate principal stress ratio or loading direction.

Fig. 15 shows the evolution of yield surfaces at very large strain for torsional shear tests on Toyoura sand. Because the initial fabric is misaligned with the loading direction, the yield surfaces (solid lines) are rotated and distorted compared to the coaxial loading case ($\alpha = 0^\circ$) at the initial state. As shearing proceeds, the fabric anisotropy variable A increases, and the yield surfaces gradually evolve towards the critical yield surfaces (dashed lines), indicating that the fabric is fully aligned with the loading direction.

It is evident that the strength of Toyoura sand is highly dependent on fabric anisotropy, which is strongly influenced by the loading direction. For triaxial compression tests from Fig. 15(a)–(d), triaxial extension tests from Fig. 15(e)–(g), and true triaxial tests with $b = 0.5$ from Fig. 15(h)–(l), the differences between the initial yield surfaces (solid lines) and the yield surfaces at the critical state (dashed lines) become more significant as α increases from 0° to 90° . The degree of the misalignment between the initial fabric and the loading direction

increases with α , leading to softer and more contractive responses. For true triaxial tests with $\alpha = 45^\circ$ from Fig. 15(m)–(p), the initial yield surfaces are the same for different b values because the loading direction is fixed at 45° . However, the strength of the sand in the σ_2 direction increases with the increase of b , leading to more significant differences between the initial yield surfaces and the yield surfaces at the critical state for higher b values. As a result, the sand exhibits softer, more contractive responses at higher b values than at lower b values, as shown in Fig. 11(c) and (f).

The above analysis demonstrates that the mechanical response of granular materials is highly dependent on the relative orientation between the fabric and the loading direction, which can be quantified by the fabric anisotropy variable A . The stress ratio, void ratio, and fabric anisotropy converge to the same critical values at the critical state, regardless of the loading path, which is consistent with the ACST framework. The proposed anisotropic SIMSAND model can effectively capture the effect of fabric anisotropy on the mechanical response of granular materials under various loading paths, including those with different intermediate principal stress ratios and loading directions.

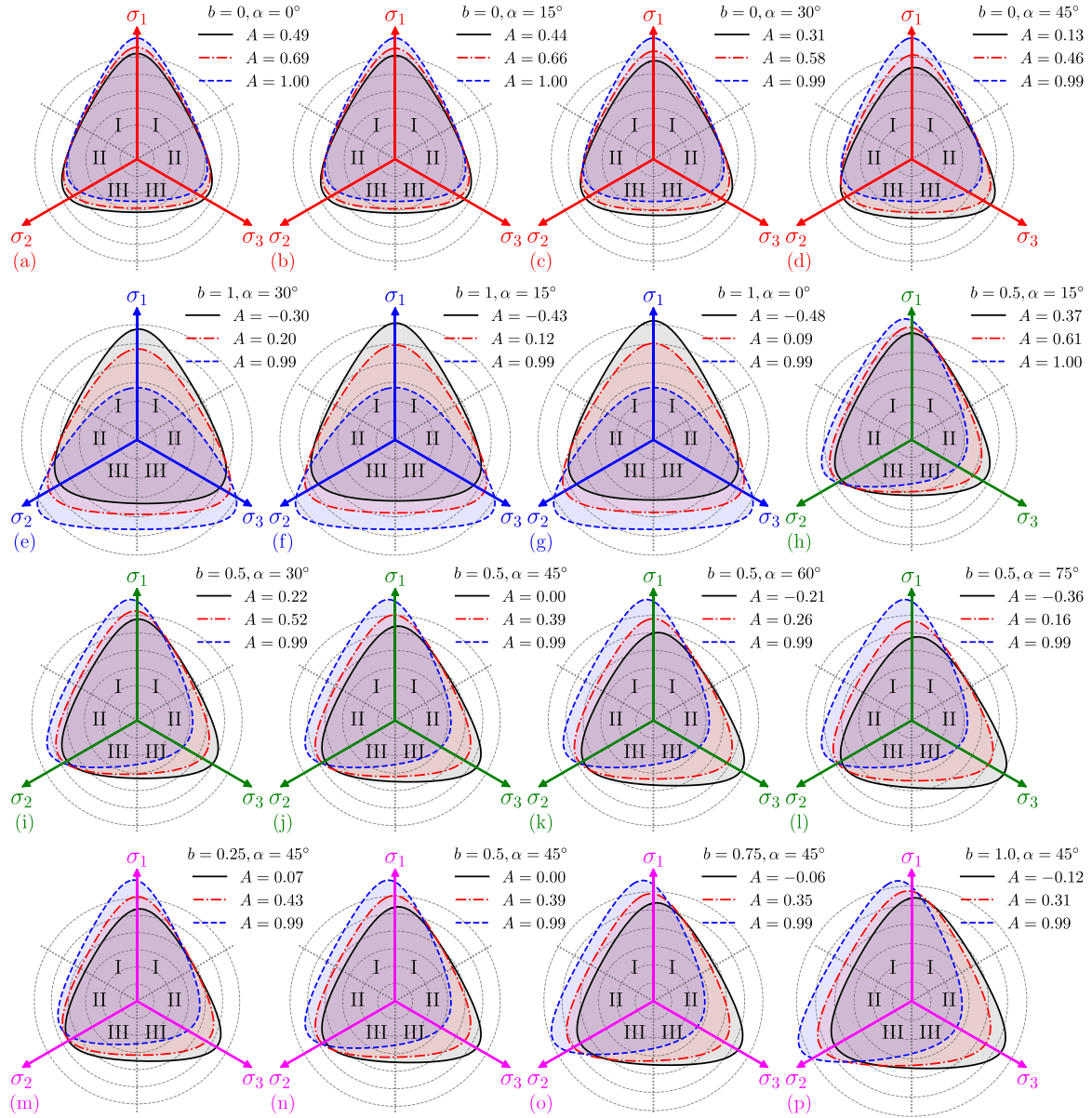


Fig. 15. Evolution of yield surfaces for torsional shear tests on Toyoura sand: (a)–(d) triaxial compression conditions with different loading directions; (e)–(g) triaxial extension conditions with different loading directions; (h)–(l) true triaxial conditions with $b = 0.5$ and different loading directions; (m)–(p) true triaxial conditions with $\alpha = 45^\circ$ and different b values.

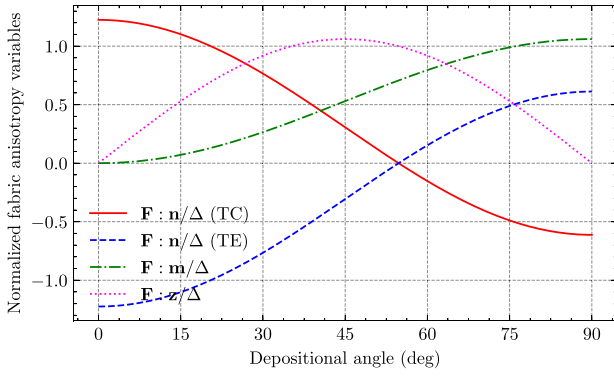


Fig. 16. Relationship between the fabric anisotropy variables and the depositional angle.

6.2. Effect of the non-proportionality between fabric tensor and loading direction

The triaxial compression and extension tests with different loading directions are simulated to investigate the effect of non-proportionality on the mechanical response of granular materials. According to Eq. (18), the fabric tensor for any given loading direction can be expressed as:

$$\mathbf{F} = \begin{bmatrix} \frac{\Delta(2 - 3\sin^2 \alpha)}{2} & 0 & -\frac{3\Delta \sin(2\alpha)}{2} \\ 0 & -\frac{\Delta}{2} & 0 \\ -\frac{3\Delta \sin(2\alpha)}{2} & 0 & \frac{\Delta(3\sin^2 \alpha - 1)}{2} \end{bmatrix} \quad (47)$$

For triaxial compression and extension tests, the loading direction $\mathbf{n}$ is given as:

$$\mathbf{n} = \begin{bmatrix} \pm\sqrt{2/3} & 0 & 0 \\ 0 & \mp\sqrt{1/6} & 0 \\ 0 & 0 & \mp\sqrt{1/6} \end{bmatrix} \quad (48)$$

where the upper sign is for triaxial compression and the lower sign is for triaxial extension. The directions of the orthogonal and non-coaxial parts of the fabric tensor can be calculated using Eq. (42), which are given as:

$$\mathbf{m} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & -\sqrt{1/2} & 0 \\ 0 & 0 & \sqrt{1/2} \end{bmatrix}, \mathbf{z} = \begin{bmatrix} 0 & 0 & -\sqrt{1/2} \\ 0 & 0 & 0 \\ -\sqrt{1/2} & 0 & 0 \end{bmatrix} \quad (49)$$

where $\mathbf{m}$ requires special derivation in Appendix I of Li and Dafalias (2020) because of the numerator and denominator in Eq. (42) being zeros. The double contractions of the fabric tensor with the directions of the proportional, orthogonal, and non-coaxial parts of the fabric tensor can be calculated as:

$$\begin{aligned} \mathbf{F} : \mathbf{n} &= \pm \frac{\sqrt{6}\Delta(2 - 3\sin^2 \alpha)}{4} \\ \mathbf{F} : \mathbf{m} &= \frac{3\sqrt{2}\Delta \sin^2 \alpha}{4} \\ \mathbf{F} : \mathbf{z} &= \frac{3\sqrt{2}\Delta \sin(2\alpha)}{4} \end{aligned} \quad (50)$$

Fig. 16 shows the relationship between the fabric anisotropy variables and the depositional angle. The figure shows that with the depositional angle increasing from 0° to 45°, $\mathbf{F} : \mathbf{n}$ decreases while $\mathbf{F} : \mathbf{m}$ and $\mathbf{F} : \mathbf{z}$ increase in TC condition. However, after the depositional angle exceeds 45°, $\mathbf{F} : \mathbf{z}$ starts to decrease while $\mathbf{F} : \mathbf{m}$ continues to increase, and $\mathbf{F} : \mathbf{n}$ continues to decrease.

It is easy to understand that $\mathbf{F} : \mathbf{n}$ decreases and $\mathbf{F} : \mathbf{m}$ increases with the increase of the depositional angle because the misalignment

between the fabric tensor and the loading direction increases with the increase of the depositional angle. Because the non-coaxial direction $\mathbf{z}$ represents normal to the π -plane of the loading direction $\mathbf{n}$ corresponding to the non-diagonal components of the fabric tensor $\mathbf{F}$ in Eq. (47), $\mathbf{F} : \mathbf{z}$ vanishes when the principal directions of the fabric tensor $\mathbf{F}$ are fully aligned with the principal directions of the loading direction $\mathbf{n}$, which occurs at a depositional angle of 0° and 90°. The maximum value of $\mathbf{F} : \mathbf{z}$ occurs at a depositional angle of 45°, where the misalignment between principal directions of the loading direction $\mathbf{n}$ and principal directions of the fabric tensor $\mathbf{F}$ is the largest.

The triaxial compression and extension tests on Toyoura sand are simulated to investigate the effect of non-proportionality on the mechanical response of granular materials based on the fabric-load coupling tensor $\mathbf{C}$ in Eq. (44), as shown in Fig. 17. The simulation results without considering the non-proportionality are also presented in Fig. 17 as dashed lines for comparison. The figure shows that the effect of non-proportionality on the stress-strain relationship is significant, especially for the case of $\alpha = 30^\circ$ and $\alpha = 45^\circ$. The non-proportionality leads to a faster evolution of the fabric anisotropy variable A and deviatoric stress towards the critical state, resulting in a stiffer and more dilative response. The effect of the non-coaxial part of the fabric tensor $\mathbf{F}$ on the stress-strain relationship is more significant than that of the orthogonal part of the fabric tensor $\mathbf{F}$ which is in agreement with Fig. 16.

Fig. 18 shows the evolution of $\mathbf{F} : \mathbf{m}$ and $\mathbf{F} : \mathbf{z}$ for the above simulations. The figure shows that $\mathbf{F} : \mathbf{m}$ and $\mathbf{F} : \mathbf{z}$ are independent on the sign of the loading direction. Upon reversal of the loading direction from $\mathbf{n}$ (TC) to $-\mathbf{n}$ (TE), $\mathbf{F} : \mathbf{m}$ and $\mathbf{F} : \mathbf{z}$ remain unchanged, which confirms the analysis presented in Dafalias and Li (2026). Both $\mathbf{F} : \mathbf{m}$ and $\mathbf{F} : \mathbf{z}$ increase with the increase of the depositional angle with the maximum values occurring at a depositional angle of 45°, consistent with the stress-strain relationship presented in Fig. 17. Both $\mathbf{F} : \mathbf{m}$ and $\mathbf{F} : \mathbf{z}$ decrease to zero at the critical state, indicating that the fabric tensor $\mathbf{F}$ is fully aligned with the loading direction $\mathbf{n}$ at the critical state.

6.3. Benchmarks

To evaluate the computational efficiency of the proposed micromechanics-inspired anisotropic SIMSAND elastoplastic model, quantitative comparisons are conducted with the micromechanical model proposed by Chang and Hicher (2005) under various loading paths. Simulation results for the micromechanical model are obtained from previous studies by Wang et al. (2025, 2026). As shown in Fig. 19, the two models are compared in terms of computational time and relative fitting error under different loading conditions. The results demonstrate that the anisotropic elastoplastic model achieves accuracy comparable to the micromechanical model while significantly reducing computational time; specifically, the anisotropic elastoplastic model's computational time ranges from approximately 1/20 to 1/1000 that of the micromechanical model, depending on the loading conditions. Additionally, the relative fitting error of the anisotropic elastoplastic model is slightly lower than that of the micromechanical model in all cases. These findings underscore the efficiency and effectiveness of the proposed micromechanics-inspired elastoplastic modeling framework in capturing the mechanical behavior of granular materials with fabric anisotropy.

7. Conclusions and future work

A novel anisotropic elastoplastic framework, inspired by a micromechanical approach, is proposed to extend existing isotropic elastoplastic models for granular materials in two steps. First, the elastic compliance tensor, which accounts for fabric anisotropy, is derived analytically from the static hypothesis, in which stress is assumed to be uniformly distributed within the granular assembly. A transformed stress tensor is introduced within the micromechanical framework to represent the

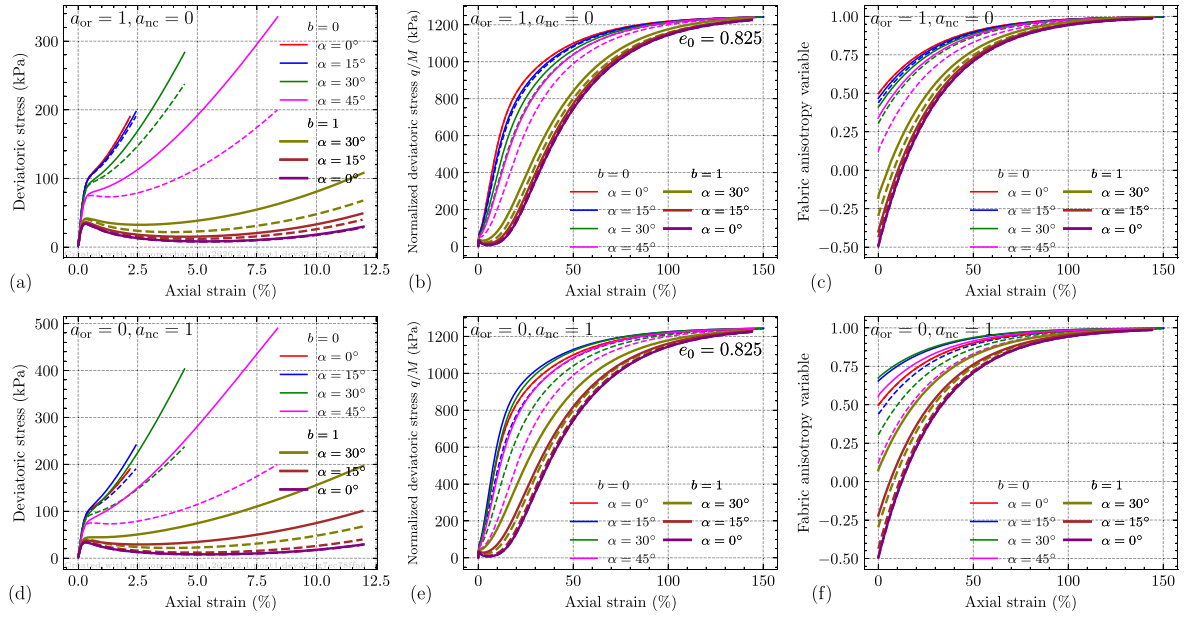


Fig. 17. Effect of non-proportionality on the stress–strain relationship, normalized shear stress at very large strains and fabric anisotropy variable: (a)–(c) considering the orthogonal part of the fabric tensor; (d)–(f) considering the non-coaxial part of the fabric tensor.

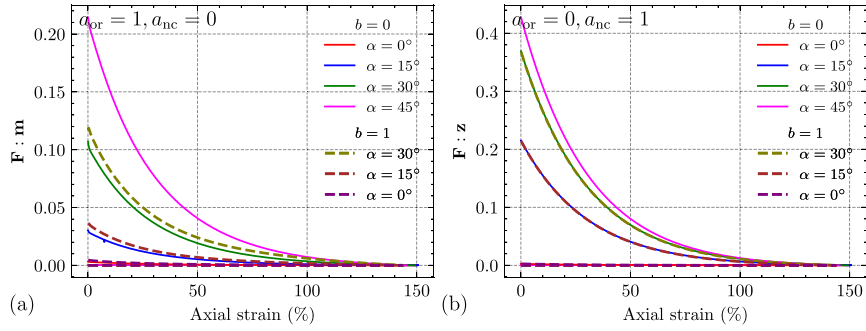


Fig. 18. Evolution of (a) $F : m$ considering the orthogonal part of the fabric tensor and (b) $F : z$ considering the non-coaxial part of the fabric tensor.

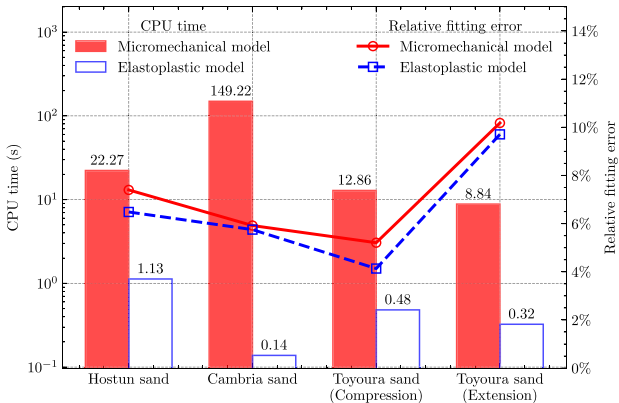


Fig. 19. Quantitative comparison of the micromechanics-inspired anisotropic elastoplastic model and the micromechanical model.

effect of fabric anisotropy on the mechanical behavior of granular materials. Utilizing the elastic compliance tensor and the transformed stress tensor, an anisotropic elastoplastic framework is established. A generalized strength criterion is incorporated to account for the effect of the intermediate principal stress. This framework enables

the generalization of conventional elastoplastic models for granular materials to include fabric anisotropy. A simple elastoplastic model is generalized using this framework, and simulations of experiments under various loading paths validate its ability to capture the effect of fabric anisotropy on the mechanical behavior of granular materials.

Further analysis reveals that the influence of fabric anisotropy on the stress–dilatancy relationship in critical-state-based elastoplastic models may violate the principles of anisotropic critical-state theory. To resolve this issue, the second step defines a fabric-load coupling tensor based on the anisotropic critical state theory, which is then used to modify the combined stress tensor, thereby revising the anisotropic elastoplastic framework. This modification ensures a gradual transition of the stress tensor from the transformed stress tensor to the original stress tensor as the fabric tensor evolves towards the critical state, eliminating the effect of fabric anisotropy on the stress–dilatancy relationship at the anisotropic critical state. The SIMSAND model is generalized using the revised framework to develop an anisotropic SIMSAND model. Validation against extensive experimental results demonstrates the model’s ability to capture the effect of fabric anisotropy on the mechanical behavior of granular materials under various loading paths. Benchmarking results indicate that the micromechanics-inspired elastoplastic model achieves accuracy comparable to the micromechanical model while significantly reducing computational cost. The proposed framework is applicable to elastoplastic models for granular materials that incorporate a stress–dilatancy relationship. Future work

$$\begin{aligned}
d\epsilon_i^t &= S_{ij}^t d\sigma_j^t \\
d\epsilon_i^t &= [d\epsilon_{11}^t, d\epsilon_{22}^t, d\epsilon_{33}^t, d\epsilon_{12}^t, d\epsilon_{21}^t, d\epsilon_{13}^t, d\epsilon_{31}^t, d\epsilon_{23}^t, d\epsilon_{32}^t]^T \\
d\sigma_j^t &= [d\sigma_{11}^t, d\sigma_{22}^t, d\sigma_{33}^t, d\sigma_{12}^t, d\sigma_{21}^t, d\sigma_{13}^t, d\sigma_{31}^t, d\sigma_{23}^t, d\sigma_{32}^t]^T \\
S_{ij}^t &= \begin{bmatrix} \frac{F_{11}^{[c_1]}}{E} & -\frac{\nu}{E} F_{33}^{[-\frac{2}{7}]} & -\frac{\nu}{E} F_{22}^{[-\frac{2}{7}]} & -\frac{6\nu}{7E} F_{12} & \frac{c_2}{E} F_{12} & -\frac{6\nu}{7E} F_{13} & \frac{c_2}{E} F_{13} & -\frac{2\nu}{7E} F_{23} & -\frac{2\nu}{7E} F_{23} \\ & \frac{F_{22}^{[c_1]}}{E} & -\frac{\nu}{E} F_{11}^{[-\frac{2}{7}]} & \frac{c_2}{E} F_{12} & -\frac{6\nu}{7E} F_{12} & -\frac{2\nu}{7E} F_{13} & -\frac{2\nu}{7E} F_{13} & -\frac{6\nu}{7E} F_{23} & \frac{c_2}{E} F_{23} \\ & & \frac{F_{33}^{[c_1]}}{E} & -\frac{2\nu}{7E} F_{12} & -\frac{2\nu}{7E} F_{12} & \frac{c_2}{E} F_{13} & -\frac{6\nu}{7E} F_{13} & \frac{c_2}{E} F_{23} & -\frac{6\nu}{7E} F_{23} \\ & & & \frac{F_{[1,2]}}{E} & -\frac{\nu}{E} F_{33}^{[-\frac{2}{7}]} & -\frac{2\nu}{7E} F_{23} & -\frac{2\nu}{7E} F_{23} & -\frac{2\nu}{7E} F_{13} & \frac{c_3}{E} F_{13} \\ & & & & \frac{F_{[2,1]}}{E} & -\frac{2\nu}{7E} F_{23} & \frac{c_3}{E} F_{23} & -\frac{2\nu}{7E} F_{13} & -\frac{2\nu}{7E} F_{13} \\ & & & & & \frac{F_{[1,3]}}{E} & -\frac{\nu}{E} F_{22}^{[-\frac{2}{7}]} & \frac{c_3}{E} F_{12} & -\frac{2\nu}{7E} F_{12} \\ & & & & & & \frac{F_{[3,1]}}{E} & -\frac{2\nu}{7E} F_{12} & -\frac{2\nu}{7E} F_{12} \\ & & & & & & & \frac{F_{[2,3]}}{E} & -\frac{\nu}{E} F_{11}^{[-\frac{2}{7}]} \\ & & & & & & & & \frac{F_{[3,2]}}{E} \end{bmatrix} \quad (51) \\
&\text{Sym.}
\end{aligned}$$

Box I.

will extend the framework to other soil models, including incremental nonlinear and hypoplastic models, and evaluate its performance in finite element analysis. Although the framework is grounded in a micromechanical approach, it does not explicitly address thermodynamic consistency. Therefore, future research will also aim to ensure that the framework satisfies the principles of thermodynamics. In addition, the current micromechanical formulation assumes particles to be equally sized spheres, which may not adequately capture the behavior of granular materials with irregular particle shapes. Although a contact-normal-based fabric tensor can be defined for irregular particles, the relationship between fabric anisotropy and the mechanical response of granular materials may differ from that observed for spherical particles. Future research will therefore investigate the influence of particle shape on the mechanical behavior of granular materials and develop a more general framework capable of accounting for particle-shape effects.

CRediT authorship contribution statement

Hai-Lin Wang: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Conceptualization. **Xiaoqiang Gu:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition, Conceptualization. **Zhen-Yu Yin:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition, Conceptualization.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used Gemini 3.1 Pro in order to improve readability and language of the paper. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix. Analytical expression of the compliance tensor

The fourth-order compliance tensor S_{ijkl}^t can be expressed as (see the Eq. (A.1) in Box I): where the constants are defined as:

$$\begin{aligned}
\nu &= \frac{3V}{2Nr^3}, E = \frac{2k_n}{rv} \frac{5\alpha}{2+3\alpha}, \nu = \frac{1-\alpha}{2+3\alpha}, G = \frac{E}{2(1+\nu)} \\
c_1 &= \frac{2(7-9\nu)}{35}, c_2 = \frac{2(7+6\nu)}{35}, c_3 = \frac{2(7+16\nu)}{35} \\
F_{ii}^{[x]} &= 1 + xF_{ii}, F_{[i,j]} = \frac{35(1+2\nu) + 2(7+16\nu)F_{ii} - 10\nu F_{jj}}{35}
\end{aligned} \quad (A.1)$$

According to the definition, the relationship between the original compliance tensor and the modified compliance tensor can be represented as:

$$S = P_e^{-1} S^t P_s \quad (A.2)$$

where P_e and P_s are the transformation matrices which are defined as:

$$\begin{aligned}
d\epsilon^t &= P_e(F) d\epsilon \\
d\sigma^t &= P_s(F) d\sigma = P_e \left\{ \frac{5}{2} \left[\left(I + \frac{2}{5} F \right)^{-1} - I \right] \right\} d\sigma \quad (A.3)
\end{aligned}$$

where $d\epsilon$ and $d\sigma$ are the original strain and stress increments, $d\epsilon^t$ and $d\sigma^t$ are the modified strain and stress increments. The transformation matrix P_e can be explicitly expressed as a function of the fabric tensor as (see the Eqs. A.5 in Box II).

Data availability

Data will be made available on request.

$$\mathbf{P}_e = \begin{bmatrix} 1 + \frac{2F_{11}}{5} & 0 & 0 & \frac{2F_{12}}{5} & 0 & \frac{2F_{13}}{5} & 0 & 0 & 0 \\ & 1 + \frac{2F_{22}}{5} & 0 & \frac{2F_{12}}{5} & 0 & 0 & \frac{2F_{23}}{5} & 0 & 0 \\ & & 1 + \frac{2F_{33}}{5} & 0 & 0 & 0 & \frac{2F_{13}}{5} & 0 & \frac{2F_{23}}{5} \\ & & & 1 + \frac{2F_{22}}{5} & 0 & \frac{2F_{23}}{5} & 0 & 0 & 0 \\ & & & & 1 + \frac{2F_{11}}{5} & 0 & \frac{2F_{13}}{5} & 0 & 0 \\ & & & & & 1 + \frac{2F_{33}}{5} & 0 & 0 & 0 \\ & & & & & & 1 + \frac{2F_{11}}{5} & 0 & \frac{2F_{12}}{5} \\ & & & & & & & 1 + \frac{2F_{33}}{5} & 0 \\ & & & & & & & & 1 + \frac{2F_{22}}{5} \end{bmatrix} \quad (\text{A.4})$$

Sym.

Since the original compliance tensor is too complex to be expressed in a concise form, it will not be provided here for the sake of brevity. However, the compliance tensor for orthotropic materials can be expressed as:

$$d\epsilon_i = S_{ij} d\sigma_j, d\epsilon_i = [d\epsilon_{11}, d\epsilon_{22}, d\epsilon_{33}, 2d\epsilon_{12}, 2d\epsilon_{31}, 2d\epsilon_{23}]^T, d\sigma_j = [d\sigma_{11}, d\sigma_{22}, d\sigma_{33}, d\sigma_{12}, d\sigma_{31}, d\sigma_{23}]^T$$

$$S_{ij} = \begin{bmatrix} \frac{1}{E} \frac{F_{11}^{[c_1]}}{(F_{11}^{[2/5]})^2} & -\frac{\nu}{E} \frac{F_{33}^{[-2/7]}}{F_{11}^{[2/5]} F_{22}^{[2/5]}} & -\frac{\nu}{E} \frac{F_{22}^{[-2/7]}}{F_{11}^{[2/5]} F_{33}^{[2/5]}} & 0 & 0 & 0 \\ & \frac{1}{E} \frac{F_{22}^{[c_1]}}{(F_{22}^{[2/5]})^2} & -\frac{\nu}{E} \frac{F_{11}^{[-2/7]}}{F_{22}^{[2/5]} F_{33}^{[2/5]}} & 0 & 0 & 0 \\ & & \frac{1}{E} \frac{F_{33}^{[c_1]}}{(F_{33}^{[2/5]})^2} & 0 & 0 & 0 \\ & & & \frac{1}{G} \frac{F_{33}^{[-1/5]}}{F_{11}^{[2/5]} F_{22}^{[2/5]}} & 0 & 0 \\ & & & & \frac{1}{G} \frac{F_{22}^{[-1/5]}}{F_{33}^{[2/5]} F_{11}^{[2/5]}} & 0 \\ & & & & & \frac{1}{G} \frac{F_{11}^{[-1/5]}}{F_{22}^{[2/5]} F_{33}^{[2/5]}} \end{bmatrix} \quad (\text{A.5})$$

Sym.

Box II.

References

- Abelev, A.V., Lade, P.V., 2004. Characterization of failure in cross-anisotropic soils. *J. Eng. Mech.* 130 (5), 599–606. [http://dx.doi.org/10.1061/\(ASCE\)0733-9399\(2004\)130:5\(599\)](http://dx.doi.org/10.1061/(ASCE)0733-9399(2004)130:5(599)).
- Amorosi, A., Rollo, F., Dafalias, Y.F., 2020. Relating elastic and plastic fabric anisotropy of clays. *Géotechnique* 71 (7), 583–593. <http://dx.doi.org/10.1680/jgeot.19.P.134>.
- Amorosi, A., Yu, Y., Yang, Z., Rollo, F., 2026. A micro-informed thermodynamically consistent plasticity model for clays accounting for double porosity and fabric. *J. Mech. Phys. Solids* 106503. <http://dx.doi.org/10.1016/j.jmps.2026.106503>.
- Argyris, J.H., Faust, G., Szimmat, J., Warnke, E.P., Willam, K.J., 1974. Recent developments in the finite element analysis of prestressed concrete reactor vessels. *Nucl. Eng. Des.* 28 (1), 42–75. [http://dx.doi.org/10.1016/0029-5493\(74\)90088-0](http://dx.doi.org/10.1016/0029-5493(74)90088-0).
- Bazant, P., Oh, B.H., 1986. Efficient numerical integration on the surface of a sphere. *ZAMM - J. Appl. Math. Mech. / Z. Angew. Math. Mech.* 66 (1), 37–49. <http://dx.doi.org/10.1002/zamm.19860660108>.
- Bignollet, F., Dormieux, L., Kondo, D., 2016. A micro-mechanical model for the plasticity of porous granular media and link with the Cam clay model. *Int. J. Plast.* 79, 259–274. <http://dx.doi.org/10.1016/j.iplas.2015.07.003>.
- Chang, C.S., Hicher, P.Y., 2005. An elasto-plastic model for granular materials with microstructural consideration. *Int. J. Solids Struct.* 42 (14), 4258–4277. <http://dx.doi.org/10.1016/j.ijsolstr.2004.09.021>.
- Chang, C.S., Hicher, P.Y., Yin, Z.Y., Kong, L.R., 2009. Elastoplastic model for clay with microstructural consideration. *J. Eng. Mech.* 135 (9), 917–931. [http://dx.doi.org/10.1061/\(ASCE\)JEM.1943-7889.0000013](http://dx.doi.org/10.1061/(ASCE)JEM.1943-7889.0000013).
- Chen, Q., Pindera, M.-J., 2020. Homogenization and localization of elastic-plastic nanoporous materials with Gurtin–Murdoch interfaces: An assessment of computational approaches. *Int. J. Plast.* 124, 42–70. <http://dx.doi.org/10.1016/j.iplas.2019.08.004>.
- Cheng, W., Yin, Z.-Y., 2024. Fractional order viscoplastic modeling of anisotropically overconsolidated clays with modified isotach viscosity. *Int. J. Plast.* 172, 103858. <http://dx.doi.org/10.1016/j.iplas.2023.103858>.
- Cheng, W., Yin, Z.-Y., 2026. A non-isothermal fractional consistency two-surface viscoplasticity model for gas hydrate-bearing sediments. *Int. J. Plast.* 196, 104547. <http://dx.doi.org/10.1016/j.iplas.2025.104547>.
- Dafalias, Y.F., Li, X.S., 2026. Modifications in anisotropic critical state theory to account for nonproportionality between fabric tensor and plastic strain rate direction. *J. Eng. Mech.* 152 (2), 06025002. <http://dx.doi.org/10.1061/JENMDT.EMENG-8795>.
- Fu, P., Dafalias, Y.F., 2015. Relationship between void- and contact normal-based fabric tensors for 2D idealized granular materials. *Int. J. Solids Struct.* 63, 68–81. <http://dx.doi.org/10.1016/j.ijsolstr.2015.02.041>.
- Ganesan, G., Mishra, A.K., 2024. A modified anisotropic hoek and brown failure criterion for transversely isotropic rocks. *Geotech. Geol. Eng.* 42 (6), 4869–4889. <http://dx.doi.org/10.1007/s10706-024-02818-0>.
- Haruyama, M., 1981. Anisotropic deformation-strength characteristics of an assembly of spherical particles under three dimensional stresses. *Soils Found.* 21 (4), 41–55. <http://dx.doi.org/10.3208/sandf1972.21.4.41>.
- Hicher, P.Y., Chang, C.S., 2007. A microstructural elastoplastic model for unsaturated granular materials. *Int. J. Solids Struct.* 44 (7), 2304–2323. <http://dx.doi.org/10.1016/j.ijsolstr.2006.07.007>.
- Khoie, A.R., Mofatteh, H., Sameti, A.R., 2023. A multiscale framework for atomistic-continuum transition in nano-powder compaction process using a cap plasticity model. *Int. J. Mech. Sci.* 255, 108482. <http://dx.doi.org/10.1016/j.ijmecsci.2023.108482>.
- Kirkgard, M., Lade, P., 1993. Anisotropic three-dimensional behavior of a normally consolidated clay. *Can. Geotech. J.* 30 (5), 848–858. <http://dx.doi.org/10.1139/t93-075>.

- Li, X.S., Dafalias, Y.F., 2004. A constitutive framework for anisotropic sand including non-proportional loading. *Géotechnique* 54 (1), 41–55. <http://dx.doi.org/10.1680/geot.2004.54.1.41>.
- Li, X.S., Dafalias, Y.F., 2012. Anisotropic critical state theory: Role of fabric. *J. Eng. Mech.* 138 (3), 263–275. [http://dx.doi.org/10.1061/\(ASCE\)EM.1943-7889.0000324](http://dx.doi.org/10.1061/(ASCE)EM.1943-7889.0000324).
- Li, X.S., Dafalias, Y.F., 2020. Noncoaxiality between two tensors with application to stress rate decomposition and fabric anisotropy variable. *J. Eng. Mech.* 146 (3), 04020004. [http://dx.doi.org/10.1061/\(ASCE\)EM.1943-7889.0001730](http://dx.doi.org/10.1061/(ASCE)EM.1943-7889.0001730).
- Liao, C.-L., Chang, T.-P., Young, D.-H., Chang, C.S., 1997. Stress-strain relationship for granular materials based on the hypothesis of best fit. *Int. J. Solids Struct.* 34 (31), 4087–4100. [http://dx.doi.org/10.1016/S0020-7683\(97\)00015-2](http://dx.doi.org/10.1016/S0020-7683(97)00015-2).
- Liao, D., Yang, Z.X., 2021. Hypoplastic modeling of anisotropic sand behavior accounting for fabric evolution under monotonic and cyclic loading. *Acta Geotech.* 16 (7), 2003–2029. <http://dx.doi.org/10.1007/s11440-020-01127-z>.
- Lou, Y., Yoon, J.W., 2017. Anisotropic ductile fracture criterion based on linear transformation. *Int. J. Plast.* 93, 3–25. <http://dx.doi.org/10.1016/j.iplas.2017.04.008>.
- Matsukawa, H., Nakai, T., 1974. Stress-deformation and strength characteristics of soil under three different principal stresses. *Proc. Jpn. Soc. Civ. Eng.* 1974 (232), 59–70. <http://dx.doi.org/10.2208/jsej1969.1974.232.59>.
- Mitaritonna, G., Amorosi, A., Cotecchia, F., 2014. Experimental investigation of the evolution of elastic stiffness anisotropy in a clayey soil. *Géotechnique* 64 (6), 463–475. <http://dx.doi.org/10.1680/geot.13.P.191>.
- Mortara, G., 2008. A new yield and failure criterion for geomaterials. *Géotechnique* 58 (2), 125–132. <http://dx.doi.org/10.1680/geot.2008.58.2.125>.
- Mortara, G., 2010. A yield criterion for isotropic and cross-anisotropic cohesive-frictional materials. *Int. J. Numer. Anal. Methods Geomech.* 34 (9), 953–977. <http://dx.doi.org/10.1002/nag.846>.
- Nicot, F., Darve, F., 2007. Basic features of plastic strains: From micro-mechanics to incrementally nonlinear models. *Int. J. Plast.* 23 (9), 1555–1588. <http://dx.doi.org/10.1016/j.iplas.2007.01.010>.
- Nicot, F., Darve, F., 2011. The h-microdirectional model: Accounting for a mesoscopic scale. *Mech. Mater.* 43 (12), 918–929. <http://dx.doi.org/10.1016/j.mechmat.2011.07.006>.
- Nicot, F., Darve, F., RNVO Group: Natural Hazards and Vulnerability of Structures, 2005. A multi-scale approach to granular materials. *Mech. Mater.* 37 (9), 980–1006. <http://dx.doi.org/10.1016/j.mechmat.2004.11.002>.
- Ochiai, H., Lade, P.V., 1983. Three-dimensional behavior of sand with anisotropic fabric. *J. Geotech. Eng.* 109 (10), 1313–1328. [http://dx.doi.org/10.1061/\(ASCE\)0733-9410\(1983\)109:10\(1313\)](http://dx.doi.org/10.1061/(ASCE)0733-9410(1983)109:10(1313)).
- Pietruszczak, S., Mroz, Z., 2000. Formulation of anisotropic failure criteria incorporating a microstructure tensor. *Comput. Geotech.* 26 (2), 105–112. [http://dx.doi.org/10.1016/S0266-352X\(99\)00034-8](http://dx.doi.org/10.1016/S0266-352X(99)00034-8).
- Reuss, A., 1929. Berechnung der fließgrenze von mischkristallen auf grund der plastizitätsbedingung für einkristalle. *ZAMM - J. Appl. Math. Mech. / Z. Angew. Math. Mech.* 9 (1), 49–58. <http://dx.doi.org/10.1002/zamm.19290090104>.
- Rollo, F., Amorosi, A., 2022. Isotropic and anisotropic elasto-plastic coupling in clays: A thermodynamic approach. *Int. J. Solids Struct.* 248, 111668. <http://dx.doi.org/10.1016/j.ijsolstr.2022.111668>.
- Shen, F., Münstermann, S., Lian, J., 2022. A unified fracture criterion considering stress state dependent transition of failure mechanisms in bcc steels at –196 °C. *Int. J. Plast.* 156, 103365. <http://dx.doi.org/10.1016/j.iplas.2022.103365>.
- Sheng, D., Sloan, S.W., Yu, H.S., 2000. Aspects of finite element implementation of critical state models. *Comput. Mech.* 26 (2), 185–196. <http://dx.doi.org/10.1007/s004660000166>.
- Stake, M., 1982. Fabric tensor in granular materials. In: IUTAM Conference on Deformation and Flow of Granular Materials, 1982. A. A. Balkema, pp. 63–68.
- Sun, Y., Huang, X., Sumelka, W., 2025. Refined SANISAND-F model for non-proportional loading: Development and application. *Int. J. Mech. Sci.* 301, 110499. <http://dx.doi.org/10.1016/j.ijmecsci.2025.110499>.
- Tobita, Y., 1988. Yield condition of anisotropic granular materials. *Soils Found.* 28 (2), 113–126. <http://dx.doi.org/10.3208/sandf1972.28.2.113>.
- Ulloa, J., Wambacq, J., Alessi, R., Samaniego, E., Degrande, G., François, S., 2022. A micromechanics-based variational phase-field model for fracture in geomaterials with brittle-tensile and compressive-ductile behavior. *J. Mech. Phys. Solids* 159, 104684. <http://dx.doi.org/10.1016/j.jmps.2021.104684>.
- Voigt, W., 1889. Über die beziehungen zwischen den beiden elastizitäts konstanten isotroper körper. *Wied. Ann.* 38, 573–587. <http://dx.doi.org/10.1002/andp.18892741206>.
- Wang, H.-L., Gu, X., Yin, Z.-Y., 2026. A generalized anisotropic failure criterion for granular materials based on a multi-scale micromechanical approach. *Int. J. Solids Struct.* 325, 113696. <http://dx.doi.org/10.1016/j.ijsolstr.2025.113696>.
- Wang, H.-L., Gu, X., Yin, Z.-Y., Zhao, C.-F., 2025. Integration of critical state theory into a micromechanical model for granular materials accounting for fabric evolution. *Comput. Geotech.* 186, 107379. <http://dx.doi.org/10.1016/j.compgeo.2025.107379>.
- Wang, H., Sun, H., Ge, X., Niu, F., 2023. An anisotropic failure criterion for cross-anisotropic soils. *KSCE J. Civ. Eng.* 27 (9), 3808–3823. <http://dx.doi.org/10.1007/s12205-023-2253-8>.
- Yang, Z., Liao, D., Xu, T., 2020. A hypoplastic model for granular soils incorporating anisotropic critical state theory. *Int. J. Numer. Anal. Methods Geomech.* 44 (6), 723–748. <http://dx.doi.org/10.1002/nag.3025>.
- Yao, Y., Lu, D., Zhou, A., Zou, B., 2004. Generalized non-linear strength theory and transformed stress space. *Sci. China Ser. E* 47 (6), 691–709. <http://dx.doi.org/10.1360/04ye0199>.
- Yao, Y., Tian, Y., Gao, Z., 2017. Anisotropic UH model for soils based on a simple transformed stress method. *Int. J. Numer. Anal. Methods Geomech.* 41 (1), 54–78. <http://dx.doi.org/10.1002/nag.2545>.
- Yin, Z.-Y., Jin, Z., Kotronis, P., Wu, Z.-X., 2018. Novel SPH SIMSAND-Based approach for modeling of granular collapse. *Int. J. Geomech.* 18 (11), 04018156. [http://dx.doi.org/10.1061/\(ASCE\)GM.1943-5622.0001255](http://dx.doi.org/10.1061/(ASCE)GM.1943-5622.0001255).
- Yin, Z.-Y., Zhao, J., Hicher, P.-Y., 2014. A micromechanics-based model for sand-silt mixtures. *Int. J. Solids Struct.* 51 (6), 1350–1363. <http://dx.doi.org/10.1016/j.ijsolstr.2013.12.027>.
- Yoshimine, M., Ishihara, K., Vargas, W., 1998. Effects of principal stress direction and intermediate principal stress on undrained shear behavior of sand. *Soils Found.* 38 (3), 179–188. <http://dx.doi.org/10.3208/sandf.38.3.179>.
- Yu, Y., Yang, Z.X., 2025. Anisotropic elasto-plastic coupling model for soils incorporating fabric evolution. *Int. J. Solids Struct.* 318, 113421. <http://dx.doi.org/10.1016/j.ijsolstr.2025.113421>.
- Zhang, Z., Soga, K., 2025. Micromechanics-inspired granular thermodynamics: A constitutive model for multidirectional cyclic shearing. *J. Mech. Phys. Solids* 200, 106170. <http://dx.doi.org/10.1016/j.jmps.2025.106170>.
- Zhao, L.-Y., Ren, L., Niu, F.-J., Lai, Y.-M., Zhu, Q.-Z., Shao, J.-F., 2023. Estimation of elastic properties and failure strength of layered rocks with a multi-scale damage approach. *Int. J. Plast.* 168, 103681. <http://dx.doi.org/10.1016/j.iplas.2023.103681>.
- Zhao, C., Salami, Y., Yin, Z.-Y., Hicher, P.-Y., 2017. A micromechanical model for unsaturated soils based on thermodynamics. In: *Poromechanics VI*. American Society of Civil Engineers, pp. 594–601. <http://dx.doi.org/10.1061/9780784480779.073>.
- Zhu, Q.-Z., Yu, Q.-J., Shao, J.-f., Zhao, X.-G., 2026. Micromechanical analysis of stress-induced anisotropic frictional damage, spontaneous localization and post-peak snapback problem in quasi-brittle geomaterials. *Int. J. Plast.* 196, 104549. <http://dx.doi.org/10.1016/j.iplas.2025.104549>.